Talking while Trans: Transfeminine Negotiations of the Cisgender Listener

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A Thesis
Presented to
The Division of Philosophy, Religion, Psychology, and Linguistics
Reed College

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Arts

Wren Arbor

May 2026

Approved for the Division
(Linguistics)

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Allison Casar

Acknowledgements

I want to thank a few people.

List of Abbreviations

CoG	Center of Gravity; a measure in Hz of mean acoustic pitch at a given moment, weighted by amplitude.
CP	Conversation Partner; the term I use to describe the cisgender woman and transgender woman who played the role of the other side of each conversation the speaker participants had.
HRT	Hormone Replacement Therapy; a common medical intervention for gender transition.
BIPoC	Black, Indiginous, and People of Color.

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Abstract

Literature Review: Research on transgender speech has primarily been of two categories: aggregated comparisons against cisgender speakers, and ethnographic exploration within queer in-groups; neither of these address how trans speakers navigate cisgender perceptions. As a non-anatomical site of gendered speech variation entrenched in hegemonic gender norms, I investigate transfeminine speakers' shifts in /s/ frontness depending on trans status of their listener to fill this gap.

Methods: I recruited 4 transfeminine speakers, and 2 woman conversation partners, one cis and one trans, acoustically comparing speakers' pronunciation of /s/ in a conversation with each partner. I additionally interviewed each speaker, assessing metalinguistic attitudes and concerns with respect to cisgender perception and hypothesizing three factors would impact shifts; feeling unsafe around cis people would encourage defensive shift towards normatively feminine /s/ pronunciations, whereas concerns about misrecognition as gay-masculine and attitudes of resistance against hegemonic femininity would encourage the opposite.

Results: Two speakers shifted towards fronter /s/ with the cis listener, and two speakers did not. The former speakers expressed concerns about misrecognition and attitudes of resistance and the latter speakers did not, counter to my predictions. All speakers expressed pressure to conform around cisgender listeners.

Discussion: Disconnects between expressed attitudes and concerns and expected speech shifting behavior point to the complexity and difficulty of sociolinguistic choices, and highlight that limitations – both external (i.e. expected to front /s/) and internal (i.e. unfamiliar with fronter /s/ pronunciations) – are equally important as agency – using linguistic features such as /s/ as strategic tools for self-definition, self-defense and resistance – in explaining transfeminine speakers' sociolinguistic situation.

Introduction

Along with the visual presentation of one's body, the sound of one's voice is among the most strongly gendered aspects of behavior. Transgender people[1] know this intimately, as our speech is highly scrutinized by those who – consciously or unconsciously – seek to enforce the gender binary (Jessee and Calder 2025). In this thesis, I investigate how this knowledge manifests by comparing English-speaking transfeminine[2] speakers' linguistic behavior across two conversational contexts: a conversation with a cisgender[3] woman, and a conversation with a trans woman. I focus on their pronunciation of the /s/[4], sound (the first sound in 'safe' and the last sound in 'face'), a well-studied site of gendered linguistic variation.

It has become widely accepted both in the academic study of gender and parts of the popular consciousness that gender is performative (Butler 1990; Henry and Hall 2025). Most simply, this means that gender does not exist outside of cultural and historical context, and is constantly constructed and reconstructed by social behavior. Under this view, scholars of gender develop their models by studying the behaviors involved in gender, and how those behaviors come to have gendering effects. Language is of primary importance to the construction of gender; speech about gender is arguably the foundation of gender's existence. For instance, the speech of medical specialists conducting an ultrasound or delivering a newborn – "it's a girl!" or "it's a boy!", or the imperatives "act ladylike" and "man up", do the performative work of enforcing and solidifying gender (Livia and Hall 1997b).

Yet, talk about gender does not represent the beginning and end of the linguistic performativity of gender. Sociolinguistics, the study of language and the social world, recognizes that variations in language use beyond the referential – less "what one says", more "how one says it" – are socially meaningful and per-

[1] I use 'transgender' or 'trans' to refer to those who identify with this label. Many transgender people understand the term broadly to include all who do not fully identify with the gender assigned to them at birth. However, the term is sometimes narrowed to refer mostly to transgender men and transgender women, who primarily identify with the opposite binary gender of the gender assigned to them at birth.

[2] I use the term 'transfeminine' to refer to thsoe who identify with this label, primarily trans women and feminine nonbinary people (especially those assigned male at birth, for whom transition is a move towards a feminine identity).

[3] I use 'cisgender' or 'cis' to refer to people who are not transgender.

[4] In this thesis, slashes denote that the enclosed text is written in the International Phonetic Alphabet. /s/ refers to the voiceless alveolar fricative speech sound. Although it looks the same as the English letter 's', this symbol refers to the sound, rather than the spelling (see International Phonetic Association 1999).

formative (Henry and Hall 2025). The common belief that one's gender can be guessed or determined from the sound of their voice attests to the importance of the sound of the voice in the construction and performance of gender. For example, if a perciever hears a voice that is higher pitched, they may assume the speaker is a woman, whereas if it is lower pitched, they may assume the speaker is a man. Here, the perciever construes high pitch to be a sign of femininity, and low pitch to be a sign of masculinity. Associations like these are known in sociolinguistics as 'indexical', as the relationship is imagined to be causal or co-occurent, much like the indexical link between smoke and fire.

It bears emphasis that the process of interpreting variations in language use (such as variations in voice pitch) as meaningful signs depends on perceivers' understanding of the world (Gal and Irvine 2019). Some indexicalities, such as the indexical link between voice pitch and gender, are the products of understandings of the world which align with and support structures of power (Calder 2021). These indexicalities come to be entrenched as expectations and norms, benefitting those who fit into gendered ideals and marginalizing those who do not. Although many cultures have somewhat overlapping gender ideals, including the idealization of the feminine voice as higher-pitched and the masculine voice as lower-pitched, the character of these indexicalities and their applications is nonetheless culturally specific. In the western, English-speaking, North American cultural context of this study, gendered indexicalities such as this one are embedded in an understanding of gender which idealizes those in power and privilege here: white, cisgender, heterosexual, middle-class men and women. These norms serve to marginalize those outside of this group, deeming their language use inadequate and inappropriate (eg. Inoue 2006; Inoue 2003; Flores and Rosa 2015; Heffernan 2023).

In investigating indexicality, existing sociolinguistic studies primarily focus on only one set of these expectations, either the normative, powerful set described above (eg. Hazenberg 2012; Podesva and Van Hofwegen 2015) and henceforth referred to as 'hegemonic', or alternative, local sets constructed within marginalized communities (eg. Mendoza 2024; Calder 2019a). Within the realm of the study of trans speech, the former type of study often interprets trans voices in their deviance from or compliance with cisnormativity, and runs the risk of tacitly enforcing hegemonic norms, marginalizing the diversity of trans voices (Calder and A. Steele 2025). Furthermore, focus within a single interpretive lens – hegemonic or otherwise – often sidelines the intersubjective nature of identity, by which identity emerges in the interaction of subjects and their establishment of contrast between self and other, speaker and listener (Bucholtz and Hall 2010). Marginalized speakers hold awareness both of hegemonic and alternative indexicalities, and must be aware of how their listener will interpret their speech.

Recent research has worked to take up this perspective, placing more even focus on both sides of the speaker-listener pair by investigating how trans voices are perceived by listeners (Jessee and Calder 2025) and trans people's understandings of these perceptions (Heffernan 2023). However, prior research has not investigated how these understandings manifest in trans speakers' actual linguistic practice. A mainstay of sociolinguistics is the study of changes within individual speakers'

language use in response to changes in context, known as style-shifting. Many of these studies investigate how speakers change their speech depending on their listener, finding listener or audience to be a primary motivating factor for shifts (e.g. Podesva 2007; Rickford and Mcnair-Knox 1994). A study of style-shifting on trans speakers treating the listener as a variable, has not yet been conducted. By filling this gap, I provide data on how transfeminine speakers navigate cisgender linguistic norms.

A primary finding of much of the existing study on listener-oriented style-shifting is that speakers often change their linguistic behavior to be more similar to the speech of their listener, known as audience design, or accomodation (eg. Rickford and Mcnair-Knox 1994). It is a truism that all speakers accomodate, but I argue, in line with up-to-date models of style shifting research (eg. Podesva 2007), that accomodation is always motivated other by social-indexical factors, and thus do not center it in my analysis. Instead, in my study of trans style shifting, I consider the motivations behind the choice to shift towards or away from a speaker in terms of how this speaker chooses to navigate cisgender norms.

Language and gender research has found that, similar to voice pitch, the English /s/ sound varies in pronunciation, and that hegemonic gender norms expect men and women to pronounce /s/ differently (Flipsen et al. 1999; Hazenberg 2012; Zimman 2013; Podesva and Van Hofwegen 2015). I focus on this sound for two primary reasons. First, unlike voice pitch, gendered variation in /s/ is not influenced by gendered anatomy; voice pitch is influenced by the size of the vocal tract, whereas /s/ pronunciation is not (Zimman 2017c). Second, research has found that when men pronounce /s/ differently than norms expect them to, their voices are percieved as sounding gay (e.g. Calder 2019a). For transfeminine speakers, this makes /s/ particularly difficult territory, as moving toward feminine /s/ pronunciations may only result in being mispercieved as gay men. In order to study how trans speakers navigate hegemonic linguistic norms, I look to this most difficult situation, investigating how transfeminine speakers shift /s/ pronunciations from a conversation with a transgender listener to one with a cisgender listener, and what factors may influence the (likely unconscious) decision to shift.

There are three possible outcome behaviors: speakers shift towards more hegemonically feminine /s/ pronunciations with the cis listener, speakers do not shift, or speakers shift away from hegemonically feminine /s/ pronunciations with the cis listener. In the first case, I predict that speakers who are unconcerned with being mistaken for gay men but who do wish to avoid being judged as insufficiently feminine by cis listeners are most likely to shift towards more hegemonically feminine /s/ pronunciations with the cis listener. The other two outcomes (either no shift or a shift away from hegemonically feminine /s/ pronunciations with the cisgender listener) would be most likely for speakers who differ from this description: those who are concerned with being mistaken for gay men, those who are unconcerned with cisgender judgements.

In order to test these predictions, I recruited 4 transfeminine speakers and 2 women conversation partners, one trans and one cis. I conducted 20-30 minute one-on-one audio recorded casual conversation sessions between each speaker and

each conversation partner in which participants were asked to collaboratively play a word game and then to chat casually and get to know each other. I then acoustically measured their /s/ pronunciations in each recording. After the conversation sessions, I interviewed each speaker, asking questions about their relationship with gender, sexuality, and their voice, with a focus on the challenges of dealing with cisgender perceptions, and assessed the extent to which speakers expressed the concerns and goals pertaining to my predictions.

I find that two speakers shift towards more hegemonically feminine /s/ pronunciations with the cisgender listener, one speaker does not shift, and one speaker shifts away from hegemonically feminine /s/ pronunciations with the cisgender listener. Counter to my predictions, the former two speakers (who shifted towards more hegemonically feminine /s/ pronunciations) were the two who expressed concern about sounding like gay men and who otherwise seemed most unconcerned with cisgender judgements. Also counter to my prediction, the latter two speakers (who did not shift towards more hegemonically feminine /s/ pronunciations with the cisgender listener) seemed concerned with cisgender judgements and did not mention concerns about sounding like gay men.

These mismatches between my predictions and the outcomes lead me to two additional conclusions. First, regardless of speakers' intentions, it is likely that speakers differ in the extent to which they (unconsciously or consciously) pick up on /s/ norms and learn to vary their /s/ pronunciations. In this way, it might be said that shifting is less accessible to some speakers than others. For trans people, this may develop and change over the course of transition; the speaker who did not shift her /s/ pronunciations was also at an earlier stage of her transition than other speakers. Additionally, trans people's experiences with voice dysphoria, the feeling that their voice does not sound the way they would like it to sound, attest to this issue of accessibility. Second, it seems that the causal order may be the reverse of my prediction with respect to misrecognition. Rather than concern with sounding like gay men causing speakers not to shift towards hegemonically feminine /s/, it seems possible that shifting towards hegemonically feminine /s/ causes speakers to be misrecognized as gay men.

These conclusions emphasize the difficulty of trans speakers' linguistic position and the injustice of hegemonic norms; in both cases, trans speakers' success in being properly percieved by cisgender speakers is limited. Trans liberation therefore cannot simply involve trans people 'improving' in their adherence to gender norms which will always exclude us, but rather must consist of the deconstruction of these norms altogether.

Chapter 1

Literature Review

1.1 Theoretical Motivations and Background

This thesis belongs to a lineage of research in the field of language and gender. Scholars of language and gender look to the gendered indexicalities of everyday linguistic behavior as a medium for gender performativity. For the bulk of this thesis, I follow the tendency of language and gender scholarship in centering indexicality as the primary framework of analysis. Although performativity is the point of departure for much of language and gender research – especially following the importance of performativity within gender studies after the foundational work of Judith Butler (1990) (see also Bucholtz 2014) – indexicality provides a number of theoretical tools, as reviewed below, which make for a more actionable lens of analysis (Henry and Hall 2025). Sociolinguists study the "social-indexical performativity of language-in-use" (Silverstein 2023, p. 17) considering performativity to be mediated by indexicality (Henry and Hall 2025). For this reason, I move forward by investigating indexicality. In this chapter, I first introduce recent approaches to indexicality in language and gender research, demonstrating the turn towards an approach which considers speaker and listener together, and underlining theoretical questions which still remain. I then review the language and gender literature on /s/, arguing that the existing scholarship is unable to provide complete answers to these theoretical questions.

Indexicality comes to sociolinguistics through semiotics, the study of meaning-making. In this thesis I adopt the model of semiotics put forth by Gal and Irvine (2019), who synthesize prior approaches, placing critical emphasis on the procedural aspect of meaning making by which phenomena come to be meaningful. In their model, individuals make meaning of their world through conjectures. Drawing upon their knowledge and understanding of the world, they recognize a phenomenon and conjecture that it may be a sign, i.e. it may be meaningful. They conjecture that the sign signals (means or represents) a particular object, its meaning. Essential to note here is that these conjectures constitute the sign-object relationship; nothing is a sign in itself, independent of a perceiver conjecturing that it is a sign and it signals an object (Gal and Irvine 2019, p. 89). Also important is

that these conjectures are based on the perceiver's knowledge and understanding of the world, and which are constructed and mediated by their cultures, communities, and individual social identities– their lived experience (Gal and Irvine 2019; see also Calder 2021). Part of a conjecture is its "grounds": the reasoning the conjecture has as to why its phenomenon is a sign and why it signals this object. A conjecture is considered to be 'indexical' if its grounds are the knowledge or belief that the sign often co-occurs with, points to, or is caused by its object (Gal and Irvine 2019, p. 96).

Sociolinguistics often focuses on finding correlations between differences in language use (eg. pronunciation, grammar, word choice) and differences in social identity, persona, or context (Eckert 2012). Particular linguistic forms are then said to index the social identities of those who use them, on the grounds of this relationship of co-occurrence (Eckert 2012). As discussed using the example of voice pitch above, if a speaker uses a particular linguistic variant, and the listener believes that form to be commonly used by those of a particular gender, they may make a conjecture on the grounds of this co-occurence, considering the use of that form to index that speaker's gender. That there are two actors in play here, a speaker and a listener, led Bucholtz and Hall (2005; 2004; 2010) to understand the indexical-performative construction of gender (and identity categories more broadly) as an intersubjective process.

Bucholtz and Hall characterize this intersubjective process as a negotiation between the indexical goals of the speaker and the indexical goals of the listener (eg. Bucholtz and Hall 2005, p. 606). Although it is possible that a listener may construe an indexical meaning from a linguistic form used by a speaker who has no knowledge of, or intent to make use of, that form's indexicality, agentivity or intentionality on the part of the speaker is common (whether conscious or unconscious). Speakers have interactional goals, and hope that their behavior will be construed as indexical of those goals, whether it be related to identity, eg. presenting feminine, or other aspects such as persona and stance, eg. seeming like a 'diva' (Podesva 2007). This agentive view arose principally in reaction to earlier sociolinguistic models which stripped speakers of agency and individuality, instead understanding their linguistic behavior simply as a product of speakers' static, essentialized identities (Eckert 2012). In practice, this turn reframes "she speaks this way because she is a woman" as "she chooses this form to index femininity." Acknowledging that individuals have choices about how to present themself and granting them authority over their self-identification is also a move towards alignment with queer and trans understandings of identity; trans people in the US often consider gender identity to be self-determined (see Zimman 2017b), rather than a consequence of bodily characteristics. Despite the value of emphasizing the speaker's goals to legitimize their identities and agency, it is nonetheless the case that the listener is the powerful other side of this negotiation.

Inoue (2006; 2003) argues that in cases where the listener occupies a privileged social position of power over the speaker, they may inhabit a privileged 'listening subject' position in which the beliefs required to uphold this social hierarchy may play a primary role in their construal of that speaker's indexical behavior and

override or erase the speaker's intentions. Studying the emergence of the notion of 'women's language' in Japan, Inoue finds that to Meiji period elite male intellectuals, schoolgirls inhabited a social position which represented a modernizing breach from traditional Japanese social structure, and a threat to their power. For this reason, when they overheard schoolgirls using the verb ending forms *teyo* and *dawa*, they construed these endings as indexical of degeneracy and social decline. Of course, schoolgirls themselves did not intend to use these forms to index a 'degenerate' identity; the purported degeneracy these forms were said to represent was a social fabrication rather than a truth about any actions these girls were involved with. In this way, male intellectual overhearers constructed a listening subject position which made negative judgmental conjectures about the indexicality of schoolgirls' speech, and constructed a fictional object for the these speech forms to signal. The male listening subject sets the narrative, erasing schoolgirl speakers' semiotic agency.

Following this important insight, research has extended theorization of the listening subject to consider race and queerness, researching white (Flores and Rosa 2015) and cisgender listening subjects (Calder 2021; Heffernan 2023; Jessee and Calder 2025; Calder and A. Steele 2025). Flores and Rosa (2015) find that when white listeners racialize speakers as non-white, they justify their position of power over those speakers by construing those speakers' speech as indexing an underspecified 'inappropriateness' for the situation (eg., the classroom), regardless of those speakers' adherence to linguistic norms for standard English, constructing a white-supremacist white listening subject which will always deem non-white speech deficient and indexical of a supposed non-white inferiority. Jessee and Calder (2025) find that cisgender listeners apply stricter gendered listening standards to an audio clip of a voice when they are told the speaker of the clip is a trans woman than when told the speaker is a cis woman, similarly situating sufficient gender performance out of reach for transgender speakers in a move to deny their identities.

Drawing on these concepts of white, cis, and male listening subjects, Calder (2021) and Calder and A. Steele (2025) raise an important critique of the sociolinguistic project, questioning which listening subjects researchers themselves may be inhabiting in their interpretation of speakers' language use. Following the conjecturing model of Gal and Irvine (2019), they note that researchers collect numerical correlation data between linguistic forms and social aspects, and use this as the knowledge on which to base their conjectures of indexical links. Incorporating the research on the listening subject, they conclude that although this data may seem objective, empirical, and scientific, purely describing the world as it is (see also Zimman 2021a), listening practices inevitably are biased by the sociocultural positioning of the researcher and the scholarship that researcher builds upon. This critique raises the question of how to research the social-indexical performativity of language-in-use without taking up and reinforcing violent listening practices and marginalizing speakers who are not white, cis, heterosexual, middle class men. Work which addresses this question employs advanced models of indexicality which encourage sociolinguistic study to move from simply asking what

meanings participants index to investigating the socio-cultural structure of individuals' indexical knowledge, and the intersubjective process of indexicality (Calder 2021; Silverstein 2003; Silverstein 2023; Barrett and Hall 2024). The concepts of the indexical field as adapted by Calder (2021), and indexical (mis)recognition as developed by Barrett and Hall (2024) have been used to model indexical knowledge and the relativity and intersubjectivity of indexicality.

The concept of the indexical field, introduced by Eckert (2008), has been adopted by Calder (2021) to capture the way in which a particular phenomenon may be conjectured to have a multiple meanings, both within a single group of people and across different groups of people. The concept of the indexical field emphasizes, connecting to Gal and Irvine's (2019) semiotic model, that indexical relationships are conjectured through listeners' knowledge. In this way, a particular linguistic form may be construed as a sign of any number of meanings, depending on which knowledge the listener uses to conjecture the relationship, and what conclusion they come to in conjecturing. Although this knowledge is mediated and constructed differently by each individual's social and cultural lived experience, it is nonetheless to a large extent shared. For example, Inoue's (2003) listening subject was not a single Meiji-era Japanese male intellectual, it was an ideological position constructed and inhabited by this male intellectual social class. Individuals who inhabit similar social and cultural backgrounds and social positions, as well as co-members of communities, are likely to draw upon a similar base of knowledge, and are likely to consider certain phenomena signs with a particular set of possible meanings. Calder (2021) conceptualizes the indexical field as this set of meanings which may be shared by a set of people with shared epistemology. By making explicit whose indexical field researchers are investigating and interpreting signs through, the fact that this interpretation is coming from particular perspectives is forefronted, preventing the universalization of the hegemonic indexical field.

While this makes progress towards a more critical and less biased move to interpreting signs, and much research has been productively done to investigate the indexical fields within particular local communities, especially community-specific systems of categorization and meaning, it is nonetheless untenable to simply split the world into non-overlapping spheres of shared knowledge and shared indexicality. Individuals bring their own indexical knowledge into interaction, and are likely aware of a number of indexical fields. The question that arises is: what happens when listeners apply an indexical field differently than a speaker intends them to, or when there is a mismatch in indexical knowledge?

Barrett and Hall (2024) model these situations using their concepts of the indexical inventory and indexical (mis)alignment/(mis)recognition. For them, an individual's indexical knowledge is structured in an indexical inventory, which is "an individual's full set of indexical fields." (p. 132) When individuals interact and engage in sociolinguistic meaning-making through shared indexical meanings, they are said to be in indexical alignment. In this case, if a speaker uses a linguistic form that they know to hold a particular indexical meaning within one of the indexical fields in their inventory, their listener would recognize (in Barrett and

Hall's term) their behavior as indexical of the meaning on which they are aligned, interpreting it in line with the speaker's intentions. However, in cases of indexical misalignment, the speaker's intended interpretation does not come through: the speaker employs a linguistic form with the intent to index a particular meaning in their inventory, but the listener does not interpret the form as indexing that meaning. Although it is possible the listener may simply be unaware of the indexicality the speaker intended to make use of, Barrett and Hall (2024) emphasize that misrecognition is often politically motivated, whether this motivation be resistance against, or reproduction of hegemonic power.

For the purposes of this thesis, it is most relevant to review the marginalizing side of misrecognition, by which misinterpretation of signs is a tool for the reproduction of power. Barrett and Hall (2024) raise the example of the transition of generic language in English from masculine (eg. *he/mankind*) forms to gender neutral forms (eg. *they/humankind*). They note that in this transition, feminists constructed a new indexical meaning around generic masculines, considering these forms an index of misogynistic views in those who use them. In reaction, anti-feminists deny this new indexicality, considering generic masculines grammatical fact and misrecognizing, or refusing to recognize, new meanings. Here, Barrett and Hall consider the anti-feminist approach to be an example of their concepts of indexical presumption – whereby those in power presume that their indexical inventory is universal and correct – and indexical denial – in which knowledge of indexicalities against one's interests are denied.

When considered together with the concept of listening subjects, indexical presumption and indexical denial emerge as tactics which privileged listening subject positions might employ to in interpreting meanings from linguistic behavior. Taken as a whole, the concepts of the indexical field, listening subjects, and tactics of indexical misrecognition do the work of, in Barrett and Hall's terms, relativising indexicality. This makes it no longer tenable for researchers of sociolinguistic indexicality to draw correlations and make their interpretations from there; instead, they must work to understand the procedure by which meaning arises and all of the factors which influence speakers' choices in language use. In particular, this thesis is interested in asking how marginalized speakers deal with these listening practices, hoping to contribute to an understanding of how speakers model listeners' indexical inventories and how they make use of these models in making (conscious or unconscious) linguistic choices?

Although all speakers may to some extent model their listeners' indexical inventories, I choose to focus on the case of marginalized speakers as a group who is under the most pressure to do so. Barrett and Hall (2024) note that indexical meanings are not evenly distributed, and I would add that the work of modeling listeners' indexical inventories is also not evenly distributed. When marginalized speakers interact outside of safe communities, they are at risk of being judged negatively by listening subjects which will presume the hegemonic indexical field and deny alternate indexicalities. Conversely, privileged speakers are unlikely to face any negative consequences due to ignorance of indexical fields belonging to marginalized groups; marginalized listeners do not have the power to inflict the

same violence as priviliged listeners might. In reaction to this situation of imbalance, marginalized speakers are coerced to protect themselves by learning hegemonic indexicalities and predicting how listeners may interpret their speech from an antagonistic listening subject position.

Heffernan (2023) reveals that transfeminine speakers are aware of the cisgender listening subject. Her study collected metalinguistic commentaries on transfeminine individuals relationships with their voices and experiences with being heard from an audio-recorded conversation she conducted between two US (South Asian American and White American) trans women and from anonymous transfeminine users on the Reddit online forum r/TransVoice. Commentaries see that cisgender listeners hear transfeminine voices through hegemonic norms and make binary judgements, categorizing voices as belonging to a man or a woman. Transfeminine speakers do not want to sound like men, and are aware of the possibility that cisgender listeners may make this judgements. In these commentaries, speakers additionally discuss what they do with this awareness and how they respond to this risk, a question I take as central to my study.

Heffernan finds that speakers simultaneously express feelings of pressure to modify their voices to avoid these negative judgements and feelings that they should not have to do so, exposing the conflict between "views of gender as self-determined, on the one hand, and of gender as intersubjectively constituted, on the other" (ii). These two approaches can be found in the commentaries from one of the audio-recording participants. This participant notes that when "interacting with the cis universe" she finds herself "deliberately modulating [her] voice" to make it "more girly," and that this is "a kind of survival tactic... to try and avoid any possibility that [she]'ll get misgendered" (23). In this way, the speaker acknowledges the risk of misgendering that comes from interacting with cisgender listeners, and her response is to change her language use in an attempt to align more closely with hegemonic feminine norms. However, she also asserts that "it's kind of just always incorrect to assume someone's gender," which and that she is uncertain about whether she would want to train herself to always use a more hegemonically feminine voice, as she wants to be able to "[think] about (.) [her] voice ... through uh- like the lens of other trans people", who would not apply the hegemonic indexical field, not assume her gender, and who would accept her voice as it is. This tension is around the decision to shift one's speech in the presence of a privileged listener who may inhabit a judgmental listening subject position is of primary interest to my study. Because she only investigated metalinguistic commentaries, Heffernan was not able to investigate how these decisions play out in transfeminine people's real interactions with cisgender listeners; this thesis aims to fill this gap.

Within the existing literature, I identify three sutdies which have considered changes in individual transgender speakers' language use based on context: Becker, Khan, and Zimman (2022), Gratton (2016), and Parnell-Mooney (2019). Becker, Khan, and Zimman (2022) find that trans men and nonbinary people on testosterone (assigned female at birth) use significantly less creaky voice (known popularly as 'vocal fry') in a reading-passage task compared to in a open-ended conver-

sation. As creaky voice is associated with femininity within the North American English hegemonic indexical field, they tentatively interpret these findings as evidence that in a more scrutinized form of speech (in which, I add, the notion of the cisgender listening subject may be more mentally present), these individuals purposefully move away from normative feminine speech in order to assert their identities as non-women.

This aligns with the findings in a study on a Toronto nonbinary friend group conducted by Gratton (2016). Gratton finds that variation between *-ing* and *in'* (eg. *working* vs. *workin'*) was more even in queer-safe spaces, where speakers used roughly the the same amount of both, and more differentiated in public, where speakers preferred one form over the other. In the hegemonic indexical field, *-ing* variant is hegemonically associated with femininity and *in'* with masculinity. Gratton finds that in public, speakers avoid the variant which is hegemonically indexical of their assigned gender at birth, understanding that cisgender overhearers are invested imposing that gender onto them and choosing to exaggerate their difference from that gender to avoid that perception. In queer-safe spaces, (in which, I add, the cisgender listening subject is not present), speakers were free to speak more neutrally and without regard for binary norms – Gratton notes that one speaker assigned male at birth mentions feeling comfortable making occasional use of a 'manly voice' in these spaces, as they know they will not face misgendering for doing so in the absence of cisgender listeners.

Parnell-Mooney (2019) also finds this sort differentiation, finding that in conversations with cis listeners, four transmasculine and two nonbinary speakers shift to pronounce /s/ further from the type of pronunciation expected of their assigned gender at birth. The trans men and the nonbinary person assigned female at birth shift further from hegemonically feminine /s/ pronunciations with with cisgender women listeners than when interacting with cisgender men, whereas the nonbinary person assigned male at birth shifts the opposite way. Although these studies present important empirical results, they do not integrate with the theoretical material discussed above – listening subjects, indexical fields, indexical recognition, and the modeling of a listener's indexical knowledge. They also do not include transfeminine speakers, and do not explicitly compare speech to a cis listener against speech to a trans listener. Filling this gap, I compare transfeminine speakers' indexical behavior across a conversation with a cisgender woman and a transgender woman with reference to their commentaries on their own gender, their voice, and being heard as a transfeminine speaker.

For my comparison, I focus specifically on the English /s/ sound due to the richness of existing research on its indexicalities and its use by queer and trans speakers, especially in connection to the sophisticated theories of indexicality discussed above. The remainder of this section introduces the literature on /s/ and reviews its connections to the concerns of transgender language and indexicality research.

1.2 The Sociophonetics of /s/

By varying the tongue position with which an /s/ is pronounced, a speaker can vary the frequencies (or pitches) of the sound produced; when the tongue position is fronter (closer to the teeth) the resulting /s/ has primarily higher frequencies, when the tongue position is backer (further from the teeth), the resulting /s/ has primarily lower frequencies (Fuchs and Toda 2010). A common measure of this frontness is the spectral center of gravity (CoG), which is the mean of all of the frequencies of an /s/ sound at a given moment in time, weighted for their amplitudes (volumes), measured in Hz. It is important to note at the outset that gendered differences in /s/ frontness do not arise from any anatomical differences associated with the sex binary, unlike other gender-correlated sites of variation such as overall vocal pitch; all people with typical vocal tracts have the anatomical equipment to produce roughly the same range of /s/ pronunciations (Zimman 2017c; 2021b).

Research on white, English-speaking, cisgender heterosexual men and women in North America – which did not include non-White, non-English-speaking, or queer speakers and listeners, a point to which I will return – find robust correlations between speaker gender and /s/ pronunciation. Flipsen et al. 1999 review 21 such studies, finding that women's /s/ pronunciations are consistently fronter than men's. Listeners in this population are aware of these correlations, and judge fronter /s/ to sound feminine, and backer /s/ to sound masculine. Strand (1999) finds that when asked to to categorize audio stimuli on a spectrum of /s/ frontness as either /s/ or /ʃ/[1] (/ʃ/ is similar to a very backed /s/), they place the boundary higher when told the speaker of the audio clip is a woman than when they are told it is a man, thus exhibiting a link between /s/ frontness and percieved gender. When listeners in this population percieve a mismatch in /s/ pronunciation and the percieved gender of a voice, they judge that voice to sound gay or lesbian. A number of studies have focused on this dynamic in perception of men's voices (eg. Levon 2014; Campbell-Kibler 2011). Munson, Jefferson, and McDonald (2006), replicates Strand's experiment, finding when listeners perceive a speaker to be lesbian sounding, they place the boundary between /s/ or /ʃ/ lower, thus expecting lesbians to back /s/.

Calder (2021) points out that these earlier correlational studies often interpret indexicalities, but fail to situate these indexicalities as specific to the populations in which they are found. It is insufficient to claim that fronter /s/ pronunciations index femininity and gayness and backer /s/ pronunciations index masculinity and lesbianness without specifying for whom these indexicalities hold. These indexical links are made specifically by cisgender heterosexual white listeners, who take the behavior of their own population as the norm, and interpret others through theis norm (an act of indexical presumption). Calder (2021) thus considers indexical links from /s/ frontness to femininity and gayness to be gender norms within the hegemonic indexical field for /s/, rather than neutral and universalizable patterns.

Calder (2021) proceeds to discuss two examples of research on outside of this

[1] /ʃ/ represents the English 'sh' sound, as in the beginning of 'shirt' or the middle of 'station'

hegemonic indexical field, one investigating Black American speakers' /s/ pronunciations, and one investigating queer nonbinary drag queens'. In the first of these studies, J. Calder and King (2022) measure the speech of find that cisgender heterosexual Black speakers in Bakersfield, California /s/ pronunciations were not differentiated by speakers' gender. They highlight that white normativity does not apply here; the men and women who participated in this study considered themselves normatively gendered cisgender heterosexuals, and cannot be said to be improperly indexing their genders, or indexing queerness. Within the indexical field of speakers in this community, Calder and King argue /s/ comes to be an index of white femininity/masculinity, and thus irrelevant to (or even to be avoided by) black speakers.

Looking at racialized aspects of /s/ in perception, Maegaard and Pharao (2015) and Pharao et al. (2014) present a similar finding in a different context. In Copenhagen Dutch, /s/ has been shown to pattern and be percieved similarly to the North American white cisgender heterosexual English-speaking population. Maegaard and Pharao find that young listeners (both white and nonwhite) in only hear fronted /s/ as feminine and gay when it is within the context of a prosodic pattern associated with middle-class whiteness, 'modern' standard Copenhagen Danish speech; in the context of a prosodic pattern associated with lower-class and immigrants, 'street' Danish, listeners instead hear fronted /s/ as associated with 'gangster'-ness and masculinity. Although Maegaard and Pharao do not apply the concept of Indexical fields to their work, it can be said that /s/ has two indexical potentials within the indexical field of these listeners, and that other indexical behavior, in this case the prosodic pattern, determines which set of indexicalities speakers take up – if the speaker sounds white with the 'modern' prosodic pattern, /s/ is interpreted through white gender norms, if the speaker sounds nonwhite with the 'street' prosodic pattern, /s/ is intepreted differently.

From these examples, two points are clear. First, local or marginalized communities' indexical fields are not simply independent of hegemonic meanings, but are instead interlinked with them, qualifying and reinterpreting hegemonic meanings. Second, variables like /s/ do not work on their own, but only come to be interpretable through a broader semiotic context; listeners in Maegaard and Pharao's studies could only interpret /s/ through its combination with prosodic types. In the second of the studies mentioned above, Calder (2019a) applies the theories of indexical orders developed by Silverstein (2023; 2003), and fractal recursivity developed by Irvine and Gal (2000) and Gal and Irvine (1995) in order to investigate this process.

As linguistic forms are construed as indexical signs and gain meaning within a particular indexical field, it is possible that new meanings will arise which build off of and are informed by the original meaning. This can take place both within a single indexical field, and cross-cutting the overlaps of multiple. The development of an indexical link between a linguistic form and a social meaning always already emerges informed by an uncountable chain of previous indexicalities and meaning-making, and for this reason is said to exist at the nth indexical order (Silverstein 2003). In the case of /s/, Calder understands the indexical link between the gender

binary and /s/ frontness to exist at the nth order within the hegemonic indexical field. When cisgender heterosexual white listeners' perception of speakers' gender does not match with their perception of that speaker's /s/ pronunciation, this recontextualization of /s/ usage forms the $n+1$st indexical order, in which a new interpretation must emerge. Here, the meaning of /s/ comes to be qualified by other construals of meaning, in that a listener's perception of a speaker's gender must be based upon other indexical conjecturing. That cisgender heterosexual white listeners construe this percieved mismatch in speaker gender and /s/ pronunciation as as gay/lesbianness evidences an erasure of transgender, nonbinary, and other queer identities; gender is assumed to be fixed, and all gender nonconformance is assumed to be gay or lesbian. Without these cisgender assumptions, the $n+1$st indexical meaning of a mismatch of percieved speaker gender and /s/ would simply be gender nonconformance, or perhaps that the speaker's gender was percieved incorrectly. With these assumptions, it comes to be gay/lesbianness. It is important to note that although the nth and $n+1$st indexicalities of /s/ are situated at different orders, they nonetheless inform and construct each other, and do not form a chronological order despite their ordered structure.

Within the San Francisco drag communities Calder (2019a) studies, an additional indexical order of /s/ is forged at the $n+1+1$nd level which is local to these communities' indexical fields, escaping the hegemonic field. A drag community Calder studies in the South of Market (SoMa) neighborhood of San Fransisco figures itself as more radically queer than another drag community in the Castro neighborhood. This is reflected by the differences in identities between these communities; SoMa queens generally identify as beyond the gender binary, many of them nonbinary or transgender, whereas the queens of the Castro neighborhood primarily identify as cisgender gay men. Calder finds that SoMa queens front /s/ more than Castro queens, and that this forges a new local indexicality to 'fierceness', anti-normativity, and and in-group status at the at the $n+1+1$nd level. Their diagram of this structure of meanings is included in figure 1.1. This emerges as an example of indexical orders cross-cutting indexical fields, where meanings in the hegemonic cisgender heterosexual white indexical field came to inform the creation of new meanings in the local, alternative indexical field.

As mentioned above, within Calder's model fronting /s/ emerges as deviant from gender norms only in those whose gender is understood as 'male' within the cisnormative conception of gender as bodily characteristics. It is perhaps surprising, then, that nonbinary and transfeminine speakers in the SoMa drag community, who would not like to be percieved as men, front /s/ exactly for the purpose of gender defiance, rather than gender conformance. It is helpful here to introduce a theoretical tool Calder employs to understand the interlinking of identity categories at these different levels of sex at birth, gender identity, sexuality, and trans/queerness– the semiotic process of fractal recursivity, as theorized by Irvine and Gal (2000) and Gal and Irvine (1995).

In fractal recursitivity, a semiotic opposition on one level becomes nested into itself, and this process can happen any number of times (Irvine and Gal 2000; Gal and Irvine 1995). In moving from the nth indexical order of /s/ to the $n+1$st (both

of which are within the hegemonic indexical field), the gender binary is nested into itself, creating a metaphorical structure where men and women are split into binary gender oppositions. This can be understood as distinguishing some notion of 'men who are men' from 'men who are women', and 'women who are women' from 'women who are men'. These become the metaphorical representation of gay men and lesbian women, since members of these groups are still fundamentally understood as types of men and types of women, relying on the aforementioned erasure of trans people and those beyond the binary. Is it then the case that by situating the distinction between cisgender gay Castro queens and nonbinary and transgender queer SoMa queens as nesting into the 'men who are women' branch, Calder as the analyst, and possibly the SoMa queens themselves, tacitly consider these queens as nonetheless a 'type of man/male'? I argue it is not. Instead, in my reading of Calder's work, SoMa queens reinterpret at the $n + 1 + 1$st level such that the fractal-recursive binary structure is broken; instead, the opposition is cisgender gay men vs. non-cisgender queer people, or as Calder articulates it, gay vs. queer. This binary still fits with the base level of /s/ indexicality, which is always present; it is not difficult for SoMa queens to figure gay men as less feminine and less subversive than queer non-cisgender people. From here, the indexicality of /s/ frontness as subversive queer non-cis femininity can be less problematically interpreted.

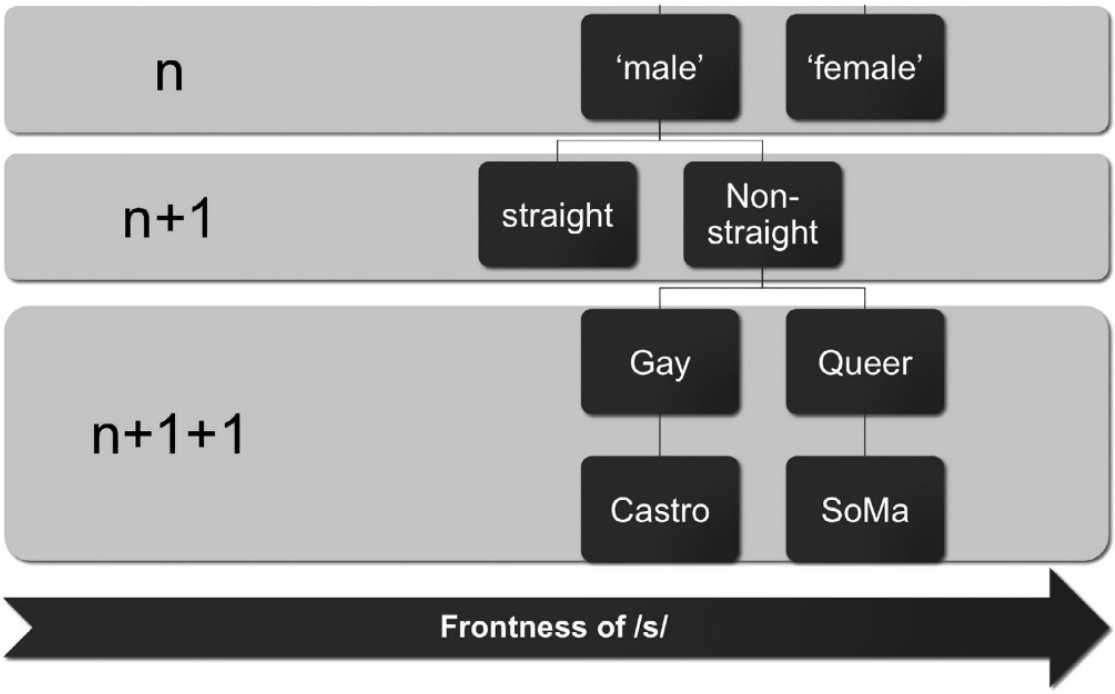


Figure 1.1: Diagram from Calder (2019a, p. 340) illustrating indexical orders of /s/ frontness and corresponding fractal recursive social oppositions across two San Fransisco drag communities.

A. J. Steele (2022) provides another example of /s/ being reinterpreted by those

outside of the gender binary. They investigate a racially-diverse community of Columbus, Ohio nonbinary individuals, finding that for Black speakers, fronted /s/ seemed to have the opposite indexical meaning, correlating positively with speakers' self-rating of their masculinity. For white speakers in this community, there was no correlation between /s/ pronunciations and masculinity ratings. This leads Steele to conclude that Black speakers subvert /s/ meanings not only to create a nonbinary identity and differentiate themselvelves from a cisgender norm, but also to repurpose and subvert a white norm. For white speakers, this dimension is not present, so the behavior is different. These patterns suggest that nonbinary speakers in this community forge further indexical orders which allow them to negotiate their specific racial and gender positions against white and cisgender normativities.

The studies discussed above illustrate the ability of speakers outside of cisgender heterosexual white populations to construct alternative and subversive meaning structures for /s/ which allow an interpretation of their unique speech practices as innovative and subversive, rather than inadequate and deviant. However, due to the dominance of the hegemonic indexical field, and the practicalities of social life, it is nonetheless the case that marginalized speakers outside of cisgender heterosexual white populations have to deal with these norms. Although it is important to note that these all speakers are socialized to believe in these norms, even if these norms will marginalize them, I focus on the most certain case, in which marginalized speakers interact with those these norms directly center and privilige.

That the hegemonic structure of indexicality for /s/ particularly excludes trans speakers is attested to by the findings of Jessee and Calder (2025). They investigate cisgender perceptions of transgender /s/ by replicating Strand (1999), comparing the placement of the boundary between /s/ and /ʃ/ when the perceiver is told the speaker is trans vs. not told they are trans (and thus is likely to assume that they are cis). They find that unlike in Munson, Jefferson, and McDonald (2006), where cisgender heterosexual speakers expect cisgender gay men and lesbians not to conform with the hegemonically normative /s/ pronunciation for their gender, transgender speakers are expected to hyper-conform– trans women are expected to front /s/ more than cis women by all listeners, and listeners who do not know a trans person also expect trans men to back /s/ more than cis men. Although indexicalities certainly would have changed in the 20 years between these two studies, it is nonetheless surprising that cisgender heterosexual white listener holds different expectations and perceptions for trans people than for cisgender gay men and lesbians, especially since it is counter to trans speakers' actual behavior; Podesva and Van Hofwegen (2015), Hazenberg (2012), and Zimman (2021b) find that trans speakers' /s/ pronunciations tend to be closer to the center of the spectrum, not as strongly gender-differentiated than cis speakers'.

Jessee and Calder (2025) analyze this through the lens of the cisgender listening subject, seeing that rather than basing perceptions on transgender speakers' actual speech, "the cisgender listening subject, which expects voices to conform to normative standards of gender appropriateness, places even stronger standards of

gender normativity on trans speakers than it does on cis speakers." (9-10) Here, the cisgender listening subject will only accept an idealized highly-conforming trans person, and construes real trans speakers as inadequately performing their genders. Because we do not live up to this idealized standard, cis percievers will misgender trans speakers, inevitably hearing them as the gender they were assigned at birth, rather than their actual gender. In combination with the $n + 1$st order of meanings for /s/, this means that attempts to move towards fulfillment of hegemonic gendered /s/ pronunciations are likely to be misrecognized as indexical of gay and lesbianness, rather than transfemininity and transmasculinity respectively. Trans speakers are 'damned if they do, damned if they don't' – if a trans woman does not front /s/, her transness may be questioned, and she may be heard as a man; if she fronts /s/, she may be construed only as a gay man, not a trans woman. In this difficult indexical situation, in which a trans speaker knows their cisgender listener may subject them to this hegemonic indexical field which excludes them, there are two primary options; defend oneself by moving towards adherence to these norms, or refuse to do so, resisting these norms and risking negative judgements. As mentioned above, transfeminine speakers studied by Heffernan (2023) consider both of these options, acknowledging the safety and possible gender-affirmation in taking up hegemonically normative femininity, while simultaneously not wishing to judge themselves by cisgender standards. Drawing upon all of this prior work on indexicalities of /s/, this thesis investigates whether transfeminine speakers shift towards hegemonically-feminine fronter /s/ pronunciations with cis listeners, and what motivations might determine the choice to shift or not shift in view

That individuals may change their /s/ pronunciations within interaction for gender-presentational and interactional purposes has been attested by Benesch (2022), Mendoza (2024), Saigusa (2016), Holmes-Elliott and Levon (2017), Levon and Holmes-Elliott (2013), Parnell-Mooney (2019), Calder (2019b), and Zimman (2017c). Holmes-Elliott and Levon (2017) and Levon and Holmes-Elliott (2013), in a study of within-speaker variation in /s/ pronunciation in British reality television shows, find that in one lower-class show, cisgender heterosexual women use backer /s/ in conversation with other women, whereas in a higher-class show, cisgender heterosexual women use fronter /s/ in this context. In order to explain these contradictory results, they conclude that the interlocutor's gender may not be the best predictor, as in both shows women back /s/ in speech activities considered more threatening. As discussed below in the methods chapter, the recorded conversations I conducted attempted to control for this stance-based shifting through the structure of two collaborative-valenced speech activities.

Moving to investigate /s/ shifting in queer speakers, Calder (2019b) finds that as SoMa drag queens get into their drag, visually transforming into their drag persona, their /s/ pronunciations get progressively fronter to index their change in persona. Benesch (2022) finds that among genderfluid people, who experience different genders at different times, shifts in /s/ index these shifts in gender, however they do not necessarily correlate with typical scales of gender such as masculinity vs. femininity, but rather with personal, alternative scales such as 'flannel' vs. 'glitter'. Mendoza (2024) presents a case study of a drag queen in Wynwood, Mi-

ami, who uses fronter /s/ pronunciations when discussing the aspects of their drag taken from white, higher-class femininity, but uses backer /s/ when discussing to their more anti-normative *chonga* drag persona. *chonga* is a queer subversive Latinx femininity which may look like "unshaved body hair and beards, as well as hoop earrings and dramatic lip liner" (307). The polyvalence of this queen's drag persona is indexed through these changes in /s/. In a similar way to the findings of Calder (2019b) and A. J. Steele (2022), this queen reinterprets the indexicality of fronted /s/, using it not to index of femininity in general, but rather to index a specific drag satire of white cisgender heterosexual higher-class femininity. In this way, meanings evolve outside of the hegemonic field.

I choose to focus on /s/ for my study, rather than other linguistic forms bearing social indexicalities, because of the theoretical depth of the research that has already been conducted on /s/; as reviewed above, this research has captured a nuanced and detailed view of the hegemonic indexical field of /s/, how it is marginalizing, and how those who are marginalized by it reinterpret and construct its meanings subversively within their communities. However, in the /s/ literature, as well as in sociolinguistic research more broadly, a gap I address is the dearth of research looking specifically at situations of indexical constraint, analyzing how marginalized speakers navigate listeners' indexical fields which marginalize them. Comparing transfeminine individuals' /s/ pronunciations in conversations with another transfem, where they can adopt their own indexicalities more freely, against their /s/ pronunciations in conversations with a cis woman, is illuminating here. More broadly, this research also addresses the absence of within-speaker research on transfeminine speakers– although there have been of studies of within-speaker variation in transmasculine people (eg. Zimman 2017c; Parnell-Mooney 2019), nonbinary people (Gratton 2016), and drag queens (Calder 2019b; Mendoza 2024), there have been no such studies focused on how transfeminine speakers speak differently in differing situations. Of specific interest is the question of individuals' navigation of the $n+1$st indexical order of /s/ frontness linked to gay masculinity; whether transfeminine speakers understand this to be a possible judgement of the cisgender listening subject, and whether this is taken into account in the choice to shift /s/. In sum, this thesis pursues the research question:

> Do transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener? What factors determine this choice?

Based on the literature reviewed, I predict:

1. Speakers who express greater feelings of danger, discomfort, dysphoria, or desire to pass around cisgender people are more likely to shift;
2. Speakers who differentiate themselves from or hold attitudes of resistance against hegemonic femininity are less likely to shift;
3. and Speakers who express concern with gay-male misrecognition are less likely to shift.

These predictions are based on the literature reviewed above, primarily the structure of the indexicality /s/ within the hegemonic indexical field it has established. Because of transfeminine people's awareness of the cisgender listening subject's application of hegemonic norms, I expect that transfeminine speakers' (conscious or unconscious) decision making around /s/ shifting will be based upon their relationship with the meanings of /s/ in the hegemonic field, as well as their relationship with hegemonic feminine norms itself; if speakers wish to distance themselves from hegemonic femininity, they will be more likely to disregard the judgements of the cisgender listening subject and refuse to take up a fronter /s/ with a cisgender listener. However, if speakers feel more threatened than defiant, feeling greater pressure to conform to hegemonic femininity, they are likely to shift towards fronter /s/ pronunciations in self-defense and to match up to the hegemonic standards. Yet, if speakers are concerned with being mistaken for a gay man via their voice, they may feel that even if they would be able to benefit from indexing hegemonic femininity, that /s/ is not an available resource through which to do so.

Chapter 2

Methodology

In order to investigate whether and how transfeminine individuals change their pronunciations of /s/ based on the trans status of their conversation partner, and to be able to interpret the possible motivations behind these patterns of change, I recruited four transfeminine individuals to be speakers and two women, one cisgender and one transgender, to be conversation partners (CPs). Each speaker participated in an audio-recorded, semi-structured casual conversation session with the each of the conversation partners, then completed a demographic survey, then participated in an audio-recorded semi-structured interview with me about their transgender experience and their relationship with their voice. This order of study activities was selected to minimize priming effects, since it is possible questions asked in the survey and interview could effect subsequent linguistic behavior.

2.1 Participants

All participants were students of Reed College between the ages of 18 and 25, recruited by flyers, word-of-mouth, and emails to the Reed College student body mailing list and to individual friends-of-friends. All speakers identified as transfeminine, and both conversation partners identified as women; one identified as cisgender and one identified as transgender. Speakers are referred to by animal-name pseudonyms, and the conversation partners are referred to as 'the Cis CP' and 'the Trans CP' respectively.

2.1.1 Survey

On the survey, participants were asked to provide any and all gender, sexuality, and race labels they identify with, as well as their pronouns. With respect to gender identity, I additionally asked participants to rank on the survey how much they generally identify with femininity and masculinity on scales of 0 to 100. However, I decided not to make use of these rankings for my analysis, as most of the participants seemed to find them uncomfortable and difficult to answer meaningfully. For the participants who were not the Cis CP, the survey also asked whether the

participant was currently on hormone replacement therapy (HRT) and for how long they had been on it, whether they had received voice feminization therapy and whether they had received voice feminization surgery. All of the trans participants were on HRT, and no participants had received voice feminization therapy or voice feminization surgery. Questions were also included to understand speakers' linguistic backgrounds– what languages they speak, whether they have primarily spoken English throughout their life or have had periods of primarily speaking other languages, the region(s) in which they grew up, and whether they had received non-gender-related speech therapy. Survey questions are provided in appendix A.

2.1.2 Interviews

After conversation sessions and surveys were completed, I also conducted 20-30 minute audio-recorded semi-structured interviews with the speakers. As a starting point, I asked questions intended to elicit background on each participant's individual relationship and history with gender and sexuality beyond the labels collected in the survey. From here, I hoped to see how participants position themselves with respect to hegemonic femininity, and how this relationship may have changed and developed over the course of transition. The discussion that followed from these questions tended also to address the question of 'outness'; what this meant for each speaker, and how private their trans and queer status was. After this, I moved to asking questions about participants' relationships with their voices. Some of these questions were modeled after those used by Heffernan (2023) to investigate speakers' understandings of the cisgender listening subject, asking speakers to talk about how they feel their voice is perceived by cis listeners and whether they might speak differently around different people. I did not directly ask speakers whether they were concerned that their voice would be mistaken as a gay man's voice, instead allowing the topic to naturally arise if it was salient to the participant. I also included in this portion questions about sexuality and the voice, such as "Do you feel that your voice sounds lesbian?", in order to get a sense of the relationship sexuality may play in determining /s/ pronunciation and shifting behavior. A full list of questions is provided in appendix B. Participants' demographic information and gender background from survey responses and interviews are summarized in tables 2.1.2 and 2.1.2.

2.2 Conversations

Before the survey and interviews, speakers participated in two 20-30 minute audio-recorded conversation sessions, one with each of the conversation partners. As these recordings would be subject to acoustic analysis of /s/, they were conducted in a sound-isolated recording booth using clip-on lavalier microphones (on the speaker as well as the conversation partner) connected to a Zoom h4n recorder. Participants were told to spend the first 10 minutes of the recording time collabo-

Participant	Gender	Sexuality	Race	First Language
Mantis (she/her)	(Trans) Woman, (Trans) Girl	Queer (Prefers to date Trans men and Nonbinary People)	Asian American, Taiwanese, Japanese, Wasian	English
Alpaca (she/it)	Woman, Girl	Lesbian	White	English
Squirrel (she/her)	Transfem, Trans Girl	Bisexual (Prefers to date Fem People)	White	English
Jellyfish (she/her)	Woman	Lesbian	White	English
Trans CP (she/her)	Woman	Ace spectrum, Demisexual	East Asian, Han Chinese	Mandarin
Cis CP (she/her)	Woman	Lesbian	white	English

Table 2.1: Participants' Demographic Identity Information, from survey and interviews

ratively playing a word game of their choice (crossword, New York Times Wordle, New York Times Connections, etc.). This prompting was intended to get participants comfortable talking with each other on the recording and to take up a collaborative and friendly stance towards one another. Controlling for stance in this way was intended to isolate stance-related variation in /s/, as indexicalities between /s/ and confrontational/threatening stances have been found (see Holmes-Elliott and Levon (2017)). Speakers and CPs were asked to time themselves and stop playing the game after the first 10 minutes of the recording, and to spend the rest of the time of the recording, another 15-20 minutes, getting to know each other, trying to find things in common, and talking about shared experiences. Since Alpaca and the Trans CP already knew each other, they were asked to catch up and chat as they would normally. The rest of the speakers and conversation partners did not know each other. The second portion of the conversation was intended to elicit free-flowing casual talk while maintaining similarly friendly and comfortable stance orientation. Although there were some moments of awkwardness, conversations did not include disagreements, arguing, or other confrontational elements and primarily maintained a lighthearted and friendly tone.

Participant	Time Out	Time on HRT
Mantis	∼3yr	1y8mo
Alpaca	∼4yr	3yr9mo
Squirrel	9mo (primarily online)	3mo
Jellyfish	>3yr	∼2yr
Trans CP	4yr	3yr

Table 2.2: Participants' Transgender Background Information, from survey and interviews

2.3 Analytical Approach

In order to answer the first portion of my research question – whether transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener – I take a quantitative empirical approach. I operationalize spectral center of gravity (CoG) of /s/, measured in Hz, as the dependent variable, and conversation partner (Cis CP vs. Trans CP) as the independent variable. In order to measure CoG of /s/, I produced a time-aligned plain-English transcript of each recording. As a first pass, I used Deepgram's Nova 3 speech recognition model, specifying that it should transcribe utterances (rather than paragraphs or as one block of text) and annotate each utterance with who it was spoken by: the speaker or the conversation partner. I then used a Python script to convert the output of this model into a time-aligned transcription file format (TextGrid) usable by the linguistic acoustic analysis software Praat (Boersma and Weenik 2023). I then manually corrected the transcripts in Praat to ensure accuracy. I then used the Montreal Forced Aligner with the included English acoustic model and dictionary to automate the production of a time-aligned phonetic transcript, which would specify, for both the speaker and the conversation partner, the start and end time in the recording of each /s/ pronunciation (McAuliffe et al. 2017). From there, I used DiCanio's (2021) Praat script to measure the CoG of each of these /s/ pronunciations in each of the recordings for the speakers and conversation partners. It was important to measure not only the /s/ pronunciations produced by the speaker participants, but also those produced by the conversation partner participants, in order to determine the extent to which the speakers may have been accomodating to the /s/ pronunciations of the conversation partners (and vice versa). I used the statistical analysis software R (R Core Team 2025) to combine, statistically analyze, and visualize the data.

In determining the second half of my research question – what factors determine whether speakers shift to a fronter /s/ pronunciation with the Cis CP – I take a qualitative approach. I use interview responses to qualitatively assess the extent to which each speaker aligns with the three criteria in my predictions, quoting and discussing instances in which speakers:

1. express feelings of danger, discomfort, dysphoria, or desire to pass around cisgender people;

2. differentiate themselves from or hold attitudes of resistance against hegemonic femininity;

3. and express concern with gay-male misrecognition.

This approach is intended to encourage my analysis of speakers on their own terms, rather than simply comparing them against a white cisgender heteronormative standard which will inevitably find them somehow faulty (see Calder and A. Steele 2025; Zimman 2021a). Instead, I aim to understand the connections between speakers understandings of their voices and their vocal behavior, underlining instances of strategic deployment of semiotic resources, as well as the oppressive limitations imposed by cisnormativity.

Chapter 3

Results

	Number of /s/ tokens		Duration of Recording	
Speaker	w/ Trans CP	w/ Cis CP	w/ Trans CP	w/ Cis CP
Mantis	177	143	21m48s	14m31s
Alpaca	476	244	30m56s	21m53s
Squirrel	388	401	25m04s	29m49s
Jellyfish	444	283	26m27s	20m08s

Table 3.1: Table of number of /s/ tokens measured and duration of each recording.

	Mean /s/ CoG (Hz)			
Participant	Overall	w/ Trans CP	w/ Cis CP	p-value
Mantis	7231	6638	7964	<0.001
Alpaca	5758	5899	5483	<0.001
Squirrel	6357	6345	6369	0.8263
Jellyfish	6154	6039	6334	<0.01
Trans CP	5133	N/A	N/A	N/A
Cis CP	6835	N/A	N/A	N/A

Table 3.2: Table of Mean /s/ CoG used by each speaker, overall as well as separated by conversation partner (CP). p-value indicates the likelihood that the difference in means between Trans and Cis conversation partner could be a result of random chance; p-values of <0.01 indicate a statistically significant difference in means.

Table 3.1 lists the duration of each recording and the number of /s/ tokens produced by the speaker participant in each recording. Table 3.2 and figure 3.1 present an overview of participants' /s/ pronunciations. Zimman (2017c) reports that so-called 'reference' studies on white English speakers presumed to be cisgender and heterosexual report a 'typical' /s/ CoG range of 6,400Hz–8,500Hz for women and 4,000–7,000Hz for men. That only Mantis and the Cis CP fell inside

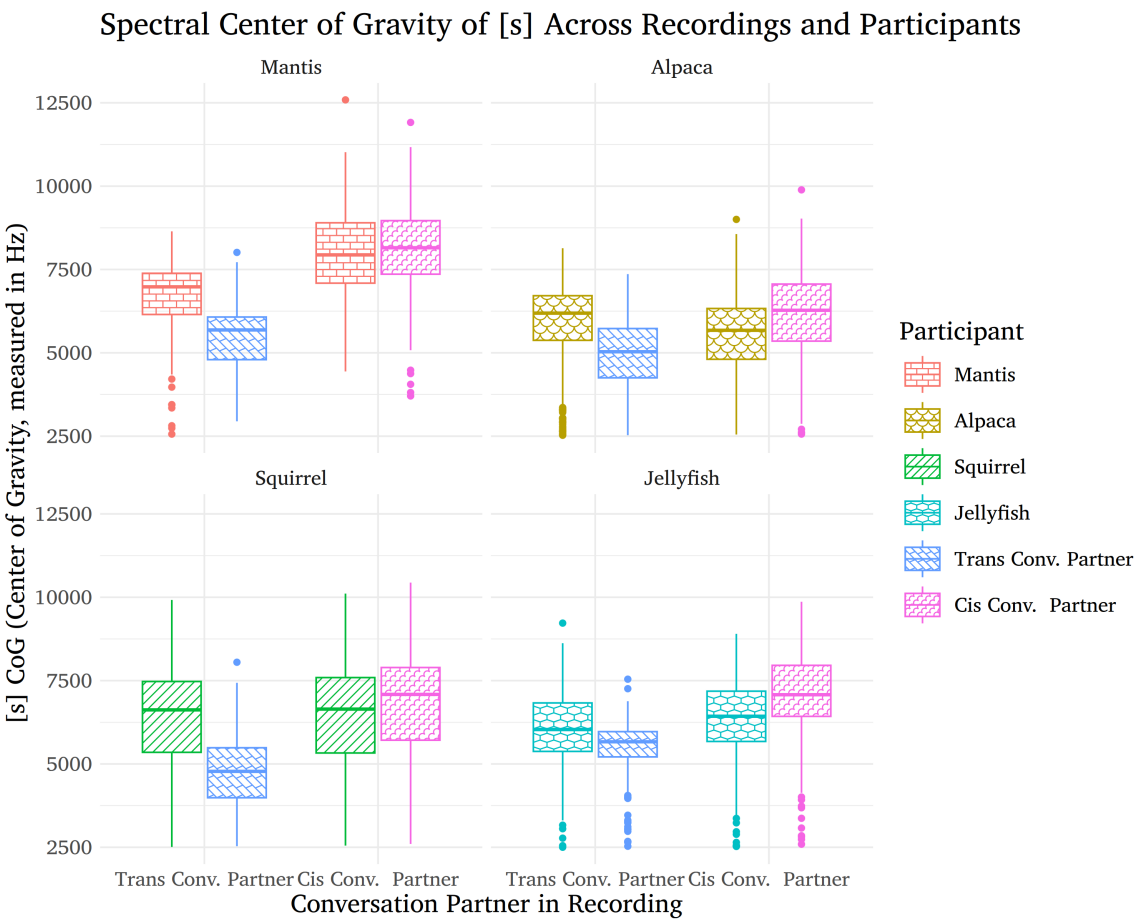


Figure 3.1: CoG of /s/ pronunciations in each recording, by each speaker.

the range expected of women, and that only Mantis fell outside the range expected of men, despite the fact that all of my participants identified as women or girls, points towards the inadequacy of these 'reference' values outside the cisgender heterosexual white population (recall that the Cis CP is a lesbian). I argue that this discrepancy should not be read as an indication of inaccurate or flawed measurement methodology, however, as my speakers' means do align with previous measurements for queer and trans women: the white trans women and cis lesbians Podesva and Van Hofwegen (2015) study in Redding, California exhibit mean /s/ CoG 6,000Hz and 6,350Hz respectively, and the white trans women Hazenberg (2012) studies in Ottawa, Ontario exhibit a mean /s/ CoG of 6,600Hz. These measures align much more closely with my speakers, with the exception of the Trans CP and Mantis, who occupy the extremes of my data. I discuss these individual cases below, but to point here is that I interpret the general correspondence of my data with previous studies of trans and queer women as confirming the validity of my measurements.

With respect to shifting, three of my four speakers, Mantis, Alpaca, and Jellyfish, exhibit statistically significant shifts in /s/ CoG from the recording with the

Trans CP to the recording with the Cis CP, and one speaker, Squirrel, did not. Of the three speakers who shifted, two, Mantis and Jellyfish, shifted in the direction of fronter /s/ pronunciations producing higher CoGs when talking to the Cis CP. This accomodation shifts their speech into closer alignment with hegemonic expectations of the feminine voice. The other speaker who shifted, Alpaca, shifted in the opposite direction, shifting to backer /s/ pronunciations with the Cis CP, moving away from the hegemonic norm in this context. With respect to the degree of shifting, Mantis exhibited the largest shift with a difference in means of >1000Hz, while Alpaca and Jellyfish shifted more subtly. Squirrel, who did not shift, maintained the same (relatively fronted) /s/ pronunciation across both conversations with a mean CoG of 6357 Hz.

In the sections that follow, I consider each speaker individually, qualitatively assessing their commentaries in the interview for the attitudinal factors listed above in section 2.3 in reference to their /s/ shifting behavior. I attempt to explain speakers' /s/ pronunciations in comparison to prior literature, my understanding of their identity and background, and these attitudinal factors. Following this, chapter 4 consists of a synthesis of these analyses, and a discussion of the implications of this data for the theories and research questions presented in chapter 1.

3.1 Speaker 1: Mantis

Figure 3.2 zooms in on Mantis' /s/ pronunciations. Her /s/ behavior is distinct from the other speakers' – her shift is the largest, and her overall mean CoG is the highest. As mentioned above, her /s/ pronunciations fall within the hegemonically-expected 'reference' range for cisgender women with an overall mean of 7231hz. In my interview with her, Mantis told me that she has "had so many people come and tell [her] that they thought [she] was a cis woman with [her] voice"; it is possible that her /s/ pitch contributes to these perceptions. She notes that these instances of having a cisgender-passing voice are the reason that she "used to worry a lot about [her] voice, but now it doesn't bother [her] as much." Here, she constructs a narrative about the development of gender confidence over the course of transition. Alpaca, Jellyfish, and the Trans CP – the other participants in the study who had been out as trans women for multiple years – all related similar stories of development in my interviews with them. Although this may seem like an expected part of a gender transition, it is nonetheless important to note, because for my speakers, part of these narratives was the development of a greater comfort with deviation from hegemonically-normative standards of femininity. Mantis notes:

> "I don't necessarily consider myself to be a woman in a way that a binary woman is a woman, but I also don't say I'm nonbinary, because I very much am like 100% woman, it's just, not in anybody else's way."

These narratives strike me as constructing a stance of resistance against the norms of unitary hegemonic femininity; in understanding her womanhood as unique to

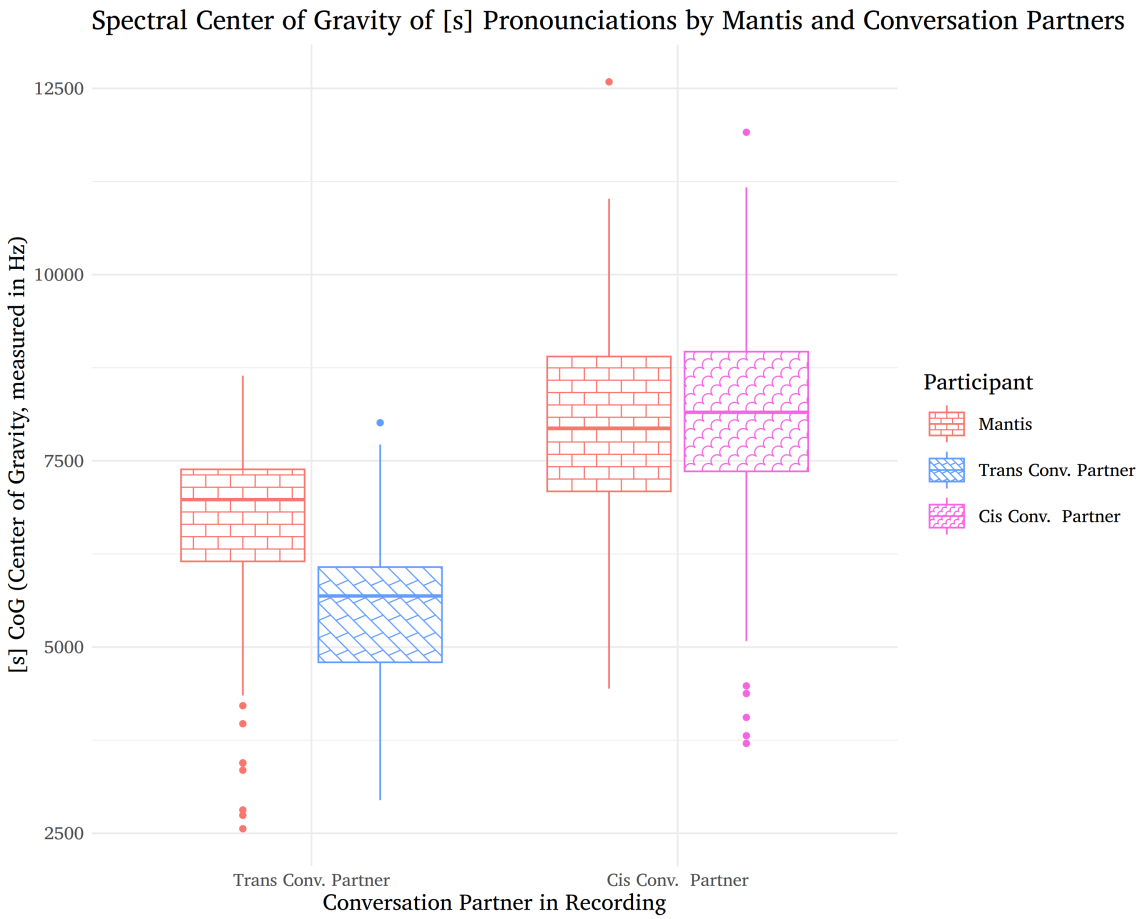


Figure 3.2: CoG of /s/ pronunciations in each recording containing Mantis, her pronunciations and the pronunciations of the conversation partners.

herself, Mantis asserts that she does not have to adhere to 'anybody else's way' in order to be '100% woman'. Here, Mantis emerges as a counter-example to one of my predictions; she shifts, despite distancing herself from cisgender norms.

An additional counter-example provided by Mantis relates to the issue of misrecognition. She does attest that sounding like a gay man is a concern, but this does not lead her to avoid fronter /s/ pronunciations with the cisgender listener as I predicted. When asked how her experience with her own voice has developed over time, she began her answer by noting that she has "always had like 'gay voice' or like feminine voice". When I asked whether her voice dysphoria was primarily the concern of not sounding feminine enough, she replied:

> "... it was, more specifically, like 'oh I just sound like a gay man'. And then there would be gay men who have gay voice ... and I'd be like 'oh I can't distinguish my voice from theirs', and it's just very frustrating, and I don't want people to think like that of me."

From this answer, it was clear that concerns about misrecognition represent an important part of Mantis's relationship with her voice. I discuss the possible in-

terpretations of this apparent mismatch in attitude and behavior in the following chapter. At this point is is sufficient to note that Mantis stands here as a counterexample to my prediction that speakers concerned with misrecognition as a gay man would avoid fronter /s/ pronunciations with the Cis CP.

On the other hand, Mantis did provide evidence to support my first prediction ,that speakers who reported greater feelings of pressure to conform or 'pass' with their voice in the presence of cisgender listeners would be more likely to shift to a fronter /s/ pronunciation with the Cis CP. In our discussion of voice dysphoria, Mantis mentions:

> "in conference sometimes ... I'm like, I don't know, maybe I shouldn't speak, because the person who just spoke before me just sounded like, so, woman, and now if I speak, people are gonna think I sound like a man. ... especially with cishet[1] women ... it's almost like a competition."

In these moments, Mantis feels that cisgender women's speech becomes the standard for feminine speech, and that the differences between her voice and the voices of cis women around her mark her speech as deficient and male. In order for feminine speech to be a 'competition', not only does there have to be a unitary way to 'win' – the cisgender norm – but further, she and her listener both have to make effort to change their speech towards this norm. This backed up in her behavior, as well as the behavior of the Cis CP; both of them shift towards fronter /s/ pronunciations with each other. In her conversation with Mantis, the Cis CP's mean /s/ CoG is 8009Hz, significantly different from her mean /s/ CoG with the other speakers (6642Hz; $p < 0.0001$). Here, the model of accomodation cannot provide a particularly clear analysis, as it is unclear who is accomodating to whom; analysis through the lens of Mantis' expressed understanding shows greater explanatory power.

3.2 Speaker 4: Jellyfish

Jellyfish, like Mantis, shifted towards fronter /s/ pronunciations when talking to the Cis CP, as shown in figure 3.3. Compared to Mantis, however, Jellyfish's upwards shift is smaller, with a difference in mean CoG of 300hz, and her overall mean CoG of at 6154hz is lower, closer to Squirrel and Alpaca. In my interview with her, Jellyfish emphasized the importance of lesbianism to her identity and her relationship with her voice, saying that "being a lesbian impacts the way that [she] move[s] through the world as a woman more than being trans does in a lot of situations," and that "[she] can sound like a boy as a lesbian, with the way that [she] present[s], and have people just be like 'oh that's like a weird dyke who smokes cigarettes'," which she considers to be an accurate description of herself. In this way, it makes sense to read her mean /s/ CoG as similar to that of the lesbian

[1] 'Cishet' is a shortened form of 'cisgender heterosexual'.

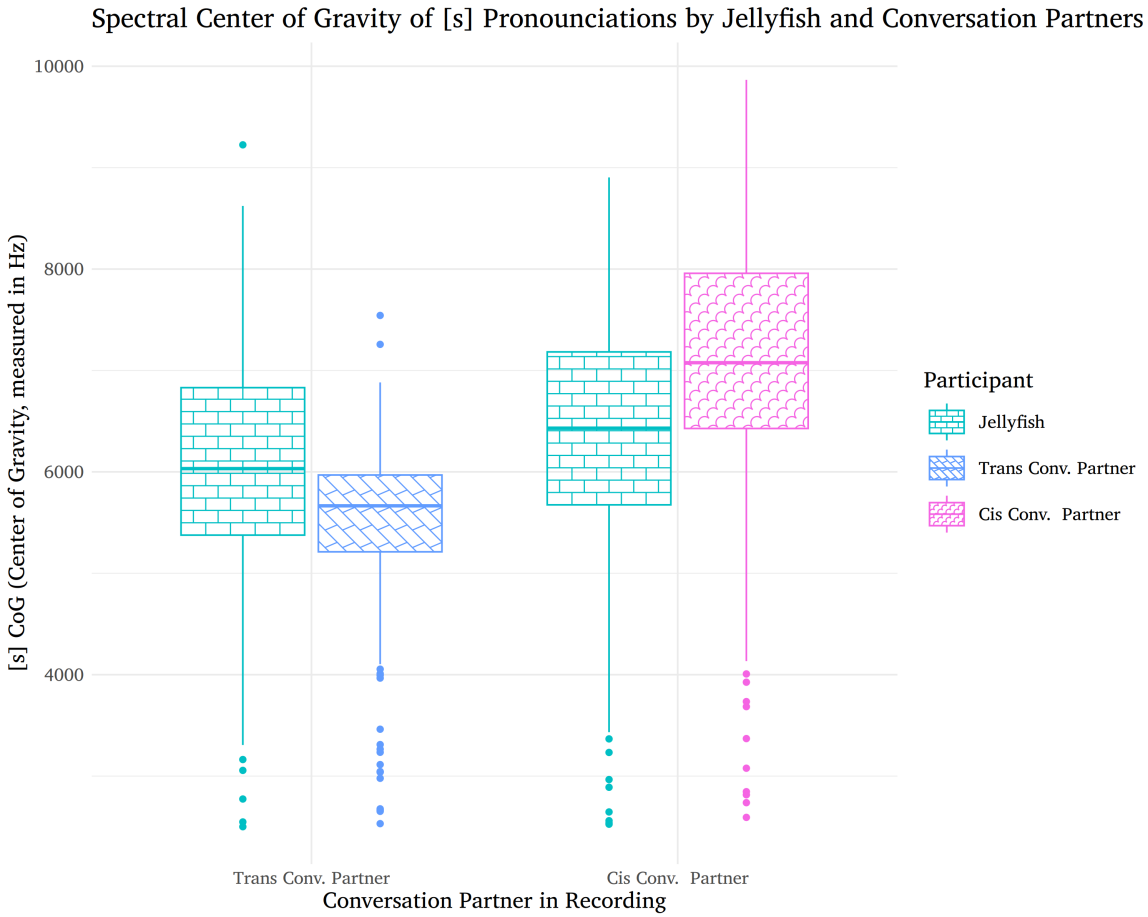


Figure 3.3: CoG of /s/ pronunciations in each recording containing Jellyfish, her pronunciations and the pronunciations of the conversation partners.

speakers Podesva and Van Hofwegen (2015) studied in Redding, California whose mean /s/ CoG was 6300Hz.

Similar to Mantis, Jellyfish lays out her transition as a process of increasing confidence in her gender and increasing ability to resist or cast off hegemonic feminine norms. For Jellyfish, this takes the shape of the development of a lesbian boy identity against earlier internalizations of ideological transphobia. She notes:

> "When I was in high school ... I would like look at all this like TERF[2] propaganda and more or less like fashion myself to be the exception to like the narrative around trans people."

She later notes that this caused her to become "kind of a psychopath, about like, presentation", paying obsessive attention at that time to "do like everything possible to fool people into thinking [she] was a cis woman". This was also the period where she notes:

[2]'TERF' is an acronym for 'trans-exclusionary radical feminist', a label for a transphobic ideology which attempts to justify itself through appropriation of feminism (see the discussion of gender-critical feminism and TERFism in Barrett and Hall 2024).

> "I realized that women whistle on their 's's way more, and I remember like consciously changing how I speak to like whistle on the 's' more."

This level of awareness of /s/ fronting is striking, and very interesting on its own. But even as an example of heightened awareness of hegemonic feminine norms and concern with having a 'passing' voice, this can be read as providing evidence for my prediction that speakers who report greater pressure to conform and pass through their voice will shift to fronter /s/ with the Cis CP. However, it is important to note that Jellyfish situates this quite clearly as an aspect of her past more than her present.

By contrast, Jellyfish's present is characterized by an attitude of resistance towards rigid conformance to a singularly feminine gender and expression. She expresses this confidently:

> " I don't think it benefits me in any sort of way to like restrict myself like that... people are way to serious about all this shit, ... like, if I'm dressed more masculine in a certain day I'll like listen to different music and I'll like talk differently and I'll move differently and I just think that's fun!"

When considered in light of my prediction that speakers who express attitudes of resistance towards hegemonic feminine norms will not shift to a fronter /s/ with the Cis CP, Jellyfish's outlook here perhaps raises the possibility a view of shifting to a fronter /s/ not as a conformist move towards the hegemonic norm, but rather as a shift in one's desired presentation, or a more neutral form of accomodation, motivated by a positive stance towards those around her. This view is also supported by her comment that she ""love/s/ talking like a boy when [she's] interacting with other boys"; here, accomodation may be read as a sort of harmonizing gender fluidity, taking up the gender of those around her.

Like Mantis, Jellyfish serves as a counter-example against my prediction that speakers who expressed concern with misrecognition would not shift /s/. Her discussion of this concern provides additional insights. She presents the issue of 'sounding like a gay guy' as particularly a concern in interaction with cis women, and as an unintentional habit, noting that:

> "It is like, annoying at times when I like, default back to talking like a gay guy around cis women ... more than sounding like a boy sounding like a gay guy is like, terrifying, because there's this relationship that, it's mostly cis women, have with like trans women where you will be accepted and tolerated as like a gay guy but anything beyond that, if you like, don't know your place more or less, you're getting into dangerous territory."

Jellyfish clarifies here that 'sounding like a gay guy' is not only a concern because of the inaccuracy and denial of her gender, but further that it represents a dangerous type of transphobia which hides itself behind an apparent friendliness towards cisgender queer people. In the phrasing 'default back to', she also calls up her

history. She mentioned in our interview that she identified as a gay boy for a period of time before realizing she was trans, and here she is perhaps indicating that 'talking like a gay guy' is a habit she developed during that time that she no longer likes but still deals with. I discuss the issue of listeners' personal history in relation to the question of gay male recognition as well as issue of intentionality and difficulty of changing how one speaks further in the next chapter.

3.3 Speaker 2: Alpaca

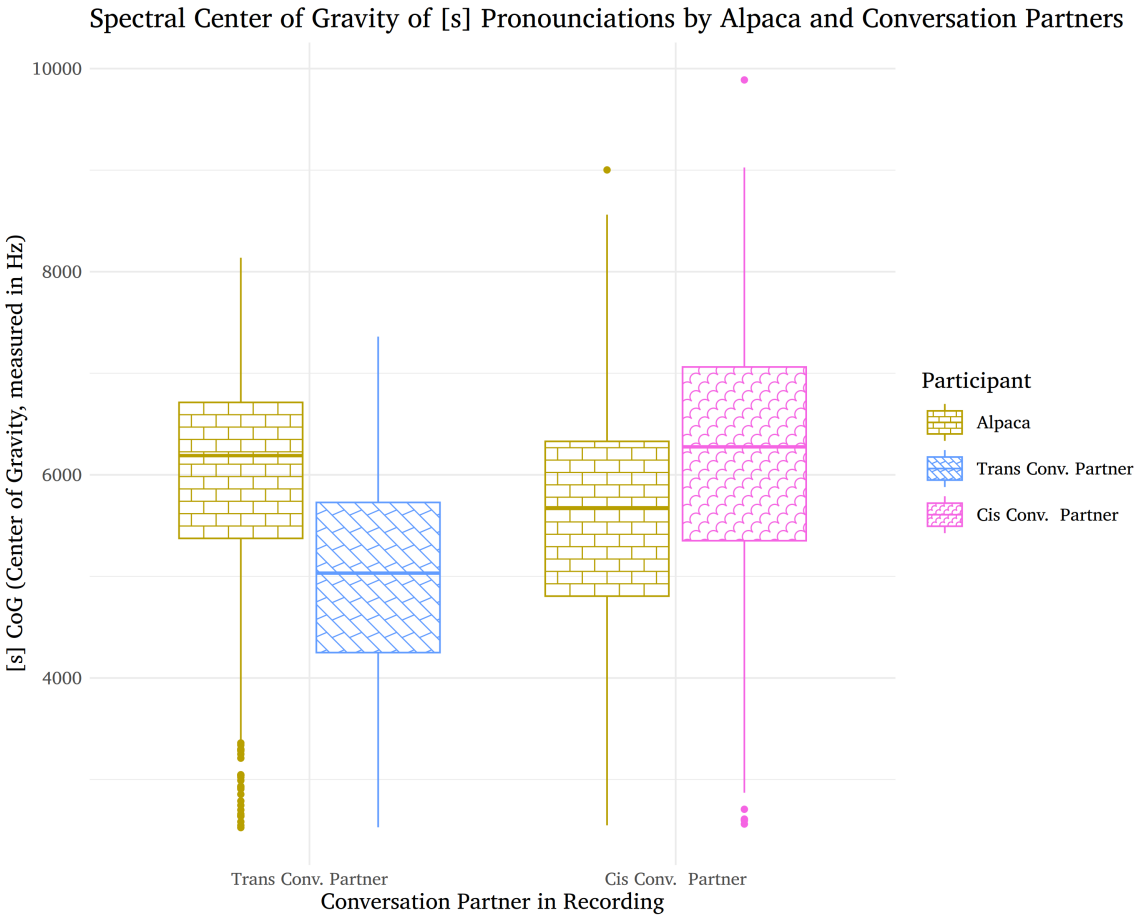


Figure 3.4: CoG of /s/ pronunciations in each recording containing Alpaca, her pronunciations and the pronunciations of the conversation partners.

Alpaca's /s/ pronunciations are shown in 3.4. Alpaca differed from Mantis and Jellyfish firstly in that she shifted her /s/ pronunciations to be backer with the Cis CP, rather than fronter; her CoG reduces by 400Hz. Also notable is that her overall average CoG of 5758Hz is the second lowest of my participants, only higher than the Trans CP's (5133Hz).

Also unlike Mantis and Jellyfish, Alpaca did not mention concern with sounding like a gay man in my interview with her. It is possible that her backer /s/

pronunciations are part of the reason that this concern does not come up for her, as her /s/ pronunciations are not front enough to trigger a misrecognizing gay-male indexicality. However, it is also possible that this concern is present, and simply went unmentioned in the interview. In this case, her shift to backer /s/ pronunciations could be to avoid this misrecognition. As either of these explanations are possible, my data on Alpaca is inconclusive with respect to the issue of misrecognition.

With respect to attitudes of resistance, the other factor my predictions associate with a lack of fronting around the Cis CP, Alpaca also differed from Mantis and Jellyfish. For Mantis and Jellyfish, the development of gender confidence involved a movement away from an initial feeling of pressure to strictly adhere to hegemonic femininity, and a gradual reclamation of behaviors which may hold hegemonic masculine associations as valid parts of a woman's identity. Alpaca describes the progression of her understanding of her gender differently:

> "I think it's been a process of just sort of like getting more comfortable with femininity, because not wanting to be masculine, is, in my thought process, is kind of different from, ... coming to terms with the fact that I do like femininity and that I am comfortable with it and that it is something I can enjoy and be comfortable in and have as my own part of my identity ..."

In this way, it seems that Alpaca's journey has been more of a gradual move towards comfort with elements of trans-affirming femininity, rather than a gradual move away from strict adherence to hegemonic femininity. Similar to the case with gay-male misrecognition, it is possible that Alpaca does distance herself from hegemonic femininity and that she simply did not mention so as explicitly as Mantis and Jellyfish. However, because my data does not include expressions of this attitude, I am unable to easily claim that she shifted away from fronter /s/ pronunciations in a move of resistance against hegemonic femininity.

With respect to the degree of concern with passing around cis people, Alpaca expressed similar feelings to the other two speakers. Although she does not word this as strongly as the prior two speakers, not associating frustration or danger with cis women's perceptions, she notes

> "when I'm around other trans people I kind of feel less pressure ... to try to present perfectly femininely, like I feel less pressure to like pass ... in class or around cis people I might try to have a more feminine voice."

Counter to my prediction, this 'pressure to pass' does not result in a shift towards fronter /s/ pronunciations with the Cis CP for Alpaca.

From the perspective of my three attitudinal predictions, I am unable to provide a solid explanation for Alpaca's shift to backer /s/ pronunciations with the Cis CP. However, there are two possible confounding factors which may contribute to an explanation. First, Alpaca notes that she was not completely sure whether the Cis

CP was cis. This was a result of my choice not to explicitly inform speakers of the trans status of the conversation partners, in order to reduce priming effects. Although she did suspect that the Cis CP may have been cis, she also believed it was possible that the Cis CP was a highly passing trans woman. As a possible confounding factor, this may also play a role in explaining Alpaca's difference in behavior from the other speakers. Second, unlike the other speaker participants, Alpaca was friends with the Cis CP, another possible confounding factor.

3.4 Speaker 3: Squirrel

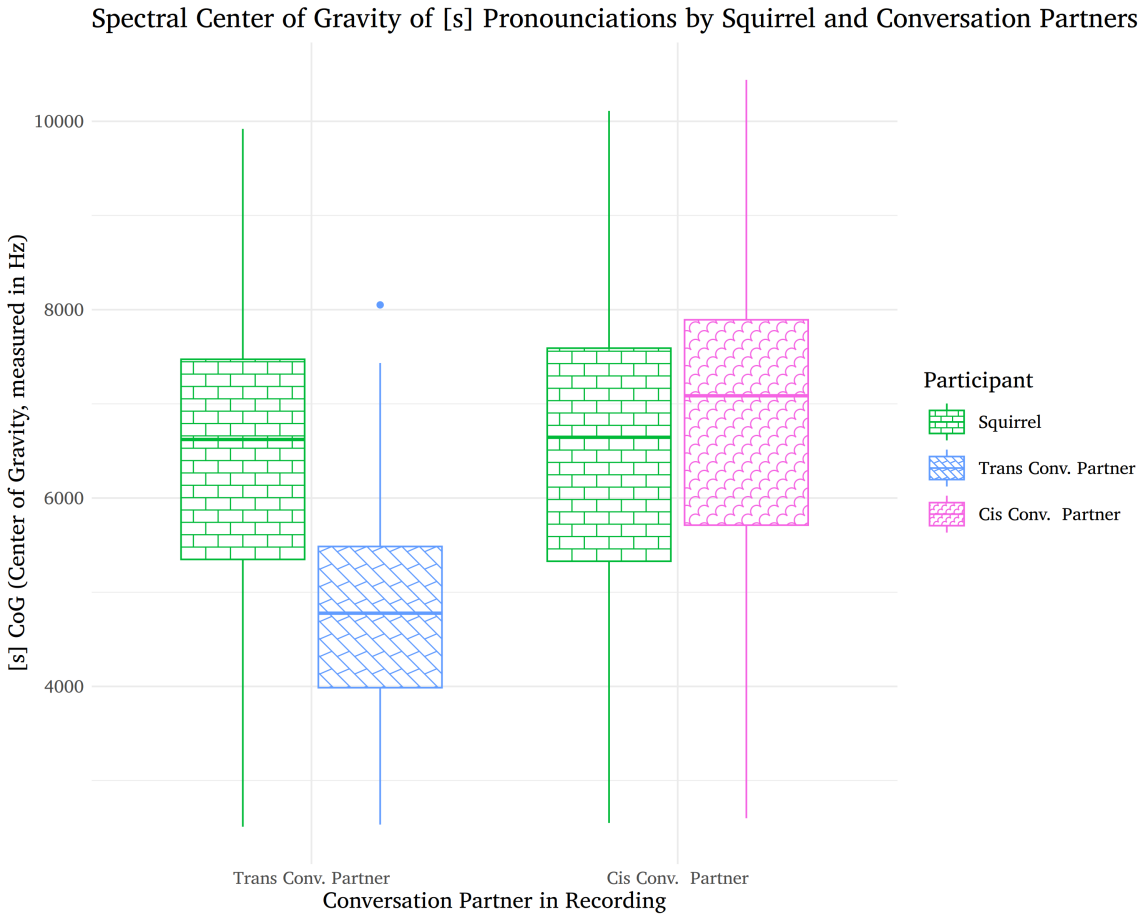


Figure 3.5: CoG of /s/ pronunciations in each recording containing Squirrel, her pronunciations and the pronunciations of the conversation partners.

Squirrel stands out as the only speaker who did not shift her /s/ pronunciations, maintaining a CoG of 6300Hz with both of the conversation partners (as shown in figure 3.5). She is also the only speaker in the early stages of her transition. She realized she was trans less than a year ago, and has not publically come out yet; only online friends and those close to her know her identity. In my interview with

her, she explained that she wished to physically transition through HRT before socially transitioning, and that she "didn't feel comfortable enough yet to" come out publically. She plans to come out next academic year, and intends to work on voice training in the summer before coming out. She did participate in this study in her transfeminine identity, and the CPs were aware that all of the people they would be talking with were transfeminine.

At this early stage of transition, it is perhaps surprising that her /s/ pronunciations are not backer, as she is the speaker participant with the second highest overall mean at 6357Hz. Although she has not yet begun efforts on voice training, it is possible she has already taken some less conscious steps towards a feminine voice, or that her /s/ pronunciation was already in this range pre-transition. In the interview, I asked her whether she felt that her voice was already feminine now, and she replied:

> "that's a complicated question... sometimes, sometimes not... when I was really young ... I used to be really insecure about my voice, but for the opposite reason, that I thought it was like too fem."

Although she cites a high overall voice pitch as the primary cause for these pre-transition insecurities, it is possible that /s/ pronunciations also played a role.

According to my predictions, speakers who hold greater attitudes of resistance towards hegemonic feminine norms and speakers who have concerns of misrecognition should be less likely to shift in /s/ pronunciations. From my interview with her, it seems that the former may be a factor in her choice not to shift. She notes that although she does intend to move towards a more feminine presentation,

> "I think a lot of times there's like this expectation of if you're transfem that you're like hyper-fem and that's not how I feel really personally, like I don't know if 'butch' is the right word for that but I definitely feel like the way that like I wanna present is not like super-hyper-fem."

Similar to Mantis and Jellyfish, Squirrel affirms here that she does not want to and does not have to strictly follow hegemonic feminine norms, and that she hopes to inhabit a femininity which may contain elements hegemonically associated with masculinity. She explains this through clothing, mentioning that she would prefer to wear all-black than a brightly colored sundress.

Squirrel does not mention concerns about being mistaken for a gay man with the sound of her voice. For Alpaca, I argued it was possible she was not concerned with gay-male misrecognition because of the backness of her /s/. This explanation is not possible for squirrel, as her /s/ pronunciations are fronter, within a similar range as reported for gay men (but also trans women) by Podesva and Van Hofwegen (2015). I discuss the implications of this mismatch in the following chapter. It is sufficient to note here that concern with misrecognition does not appear to be a factor in Squirrel's lack of shifting, counter to my predictions.

Squirrel's lack of shifting may also be interpreted as a counter-example to my prediction that speakers who express greater concern with sounding feminine and

passing would be more likely to shift, as it seemed that this concern was quite present for her. When I asked her about her relationship with her voice, she begun by explaining that in online voice chat interactions,

> "I like don't talk to a lot of people like that I would normally talk to if I was cis... as soon as I like basically came out to like my [online videogaming group]... I stopped talking entirely, on voice. ... I feel like it's easier for them to perceive me and like use correct pronouns like, if they don't hear me, because my, there's voice dysphoria, and like, my voice isn't like, they way I would want it to be if I could like pick it basically."

She explains further that

> "I feel like inherently when like you hear someone for the first time you kind of assign like gender in your head to like that person... if you hear a stereotypically more masc voice then you might like automatically start like 'he/him'ing someone in your head."

Here, she implicitly characterizes her voice in its current state as 'stereotypically more masc', in that she feels that not using her voice is the best way to avoid being perceived as a man in online interactions with cisgender people. For her, this falls under a general preference articulated as "the more ways I can like not perceived, the better." In this context, it is perhaps understandable that pressure to pass does not lead to a shift in /s/, as Squirrel perceives a passing or feminine voice as not yet in reach at her current stage of transition.

Chapter 4

Discussion

Based on the results above, I argue that the first research question of this thesis – Do transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener? – is that some speakers do. Although this may initially seem obvious in light of the statistically significant shifts, it is necessary to consider whether the speakers are specifically reacting to the trans status of the CP and the hegemonic indexical field of /s/, and are not acting by some other principle. A primary concern here is the issue of accomodation; could these shifts be explained simply as the speakers pronouncing /s/ more similar to the CP they are talking with, rather than in reaction to whether the CP is cis or trans? The discussion that follows considers this question in detail. I find that although I cannot answer this question with certainty, that I find it more likely, and more interpretively productive, that speakers are specifically reacting to the trans status of the CP and the hegemonic indexical field of /s/, rather than simply accomodating.

Due to the small scale and scope of this study, my research design is unable to fully eliminate the factor of accomodation. In order to do so, it would either need to be the case that the two CPs have the same /s/ pronunciations – in which case accomodation could not result in any shift – or that more than one of each type of CP was included (e.g. two trans CPs and two cis CPs) – in which case shifts across listeners of differing trans status could be compared with shifts within listeners of the same trans status. Future research employing these designs could be highly illuminating. In the absence of these measures, it is certain that accomodation is taking place for Mantis and Jellyfish; their mean /s/ pronunciations fall in between the mean /s/ pronunciations of the CPs, and their shifts move them closer to the pronunciations of their CP.

To point, though, is not whether these speakers are accomodating, but whether they accomodate equally to both CPs. In the latter case, the trans status of the CP does not matter, and these speakers accomodate to the two CPs equally. It is at this point that my discussion splits, as I two possible interpretations here. In the first, these speakers do not accomodate equally to both CPs. Mantis, in this view, accomodates less to the Trans CP than she does the Cis CP; with the Cis CP, the two speakers converge on extremely similar pronunciations, whereas with the

Trans CP, the standard deviations of their /s/ pronunciations do not overlap (recall 3.1). Jellyfish, in this view, is seen to accomodate to the Trans CP more than she does the Cis CP, with a smaller difference in means with the Trans CP than with the Cis CP. However, the second interpretation would argue that that these patterns are best explained through a difference in speakers' baseline /s/ pronunciation, and that speakers are accomodating equally to both CPs. In this case, Mantis's baseline would be closer to the /s/ pronounciations of the Cis CP, and Jellyfish's closer to the Trans CP's, and Mantis and Jellyfish simply accomodate within their ranges without regard to the trans status of the CP. With the data I have, I am unable to determine speakers' baselines; therefore, both interpretations are possible. I return to the first interpretation (which I ultimately prefer) below, here turning to discuss the implications of the second.

If Mantis and Jellyfish are considered accomodators, how would this view (in which the trans status of the CP and the hegemonic indexical field of /s/ are not considered relevant) distinguish them from Alpaca and Squirrel, who do not accomodate? Research on sociolinguistic accommodation generally considers differences between individuals' degree of accomodation within two frameworks, neither of which, I argue, provide clear answers here. First, based upon patterns found within the cisgender heterosexual white population, it is often assumed that women accomodate more than men (e.g. **giles_communication_2016**). As all four of my speakers identify as women or girls, this generalization would thus expect them to be similarly inclined towards accomodation, and cannot differentiate between Mantis and Jellyfish. Second, accommodation is considered as a tactic of intersubjective adequation (e.g. **giles_communication_2016**), see Bucholtz and Hall 2005 on tactics of intersubjectivity), whereby speakers accomodate in order to create a sense of similarity and solidarity between each other. This conceptualization is more helpful in distinguishing why an individual accomodates more to one CP than another than in distinguishing why some individuals accomodate more than others to the same CPs. When the CPs are considered together, it is not clear that Mantis and Jellyfish were more interested in creating solidarity between themselves and the CPs than Alpaca and Squirrel were. This might expected if, say, there was some in-group to which the CPs, Mantis, and Jellyfish belonged, but Alpaca and Squirrel didn't. I am unaware of any way in which this could be the case; Mantis and Jellyfish did not seem to have any more in common with the CPs than Alpaca and Squirrel did.

At this point, the interpretation in which Mantis and Jellyfish are viewed as accomodators, and speakers do not react to the trans status of their CPs nor the hegemonic indexical field of /s/, ceases to provide helpful interpretations. Although I cannot reject this interpretation outright – it is always possible there is factor outside of my data which could explain why Mantis and Jellyfish are simply more inclined to accomodate than Alpaca and Squirrel – I nonetheless find it unlikely that the trans status of their CPs nor the hegemonic indexical field of /s/ are not relevant to these speakers' behavior, and find including these considerations interpretively productive.

Instead, I argue that social explanations motivations must be considered.

it is necessary to evaluate whether my data can be explained through accomodation, without reference to the factor of trans status of the listener, the dynamic of the cisgender listening subject's hegemonic indexicality of /s/, or the trans background and intentions of the speakers. Is it possible to explain the differences in my speaker's behavior – that Mantis and Jellyfish shift fronter, Alpaca shifts backer, and Squirrel does not shift – in this way?

It is at this point that CP's trans status emerges as a necessary explanatory factor;

wouldn't it predict that transfems would accomodate more to other transfems than cisfems? Second, still is not able to differentiate between squirrel + alpaca and mantis + jellyfish.

Mantis and Jellyfish's shifts are read as accomodation, why did they choose to accomodate where Squirrel and Alpaca did not? If it were simply the case that all speakers accomodate, these speakers would be expected to accomodate too.

I nonetheless argue that accomodation should be considered an outcome, rather than a complete explanation. In other words, it is insufficient to state that speakers are accomodating without interpreting the motivations behind their accomodation.

I argue that both of these frameworks must be considered as indexical moves rather than interpretive explanations. In the case of the first pattern, I problematize as a generalization which can only become an interpretation when normativity is assumed to be an explanation rather than an object of criticism– it is not sufficient to say 'she accomodates because she is a woman', without questioning what populations the generalization is based upon, and why women might accomodate more than men. Here, a reinterpretation through indexicality is needed. That accomodation research within cisgender heterosexual white populations has found women to accomodate more than men can be interpreted as an indexical relationship of co-occurrence; it is possible that the act of accomodating itself is indexical of femininity within the hegemonic indexical field.

At this juncture, I argue that interpretation of my speakers' behavior must return to the social-indexical framework I set out in the literature review, confirming the relevance of trans status of the listener, the indexical structure of norms surrounding /s/ and accomodation, as well as the background and intentions of the speakers as transfems. An approach which does not incorporate these factors as relevant, simply relying on models of pure accomodation, cannot differentiate between these speakers meaningfully.

In light of these considerations, I return to the question posed above: why Jellyfish and Mantis front /s/ with the Cis CP, accomodating and moving towards the hegemonic norm, and why Squirrel and Alpaca do not. Is it possible that Jellyfish and Mantis are just accomodators, and that the cisness of the listner doesnt matter? Well, I do not have any reason to believe that they intend to index femininity or positive stances towards their listeners more than Squirrel and Alpaca, so I don't think so. Also, they don't accomodate to the same extent to the trans listener as they do to the cis listener: Mantis's competition case.

I argue she does not because she is not used to presenting feminine in face-to-face (not online) interactions, and has not yet learned (consciously or uncon-

sciously) how to pronounce /s/ differently than she did pre-transition. Zimman (2021b) demonstrates that a multi-dimensional model of gender is needed to explain the /s/ pronunciation patterns of his gender-diverse set of speakers, including gender assigned at birth, current gender identity, and gender role – what others know about a speaker's gender. My data agrees with Zimman's affirming the importance of considering gender role, since gender role likely provides an explanation for Squirrel's lack of a shift.

That Squirrel did not shift, and that Alpaca shifted in the opposite direction of accomodation

This provides an answer to the first and simplest research question of this thesis. This result – the shifts exhibited by Mantis and Jellyfish to fronter /s/ pronunciations – can be interpreted as evidence of the influence of the cisgender listening subject. In addition to its relevance to transfem thinking about the voice and being heard (Heffernan 2023) and its inhabitation by real cisgender listeners (Jessee and Calder 2025), it is clear that the presence of the cisgender listening subject does, for Mantis and Jelllyfish, impact actual linguistic behavior in the form of /s/ shifting. This study thus supports the usefulness of listening subjects as a theorization of the mediation between perception of speech and production of speech, especially for trans speakers.

Central to this mediation is the question of how speakers' mental models of how their speech may be perceived inform their subsequent choices of linguistic variants. My results complicate this question; the relationships between the attitudes and concerns expressed in the interviews and observed behavior are mostly unexpected, and did not align with the model I initially constructed. It is notable that even though all four speaker participants did express concerns about cisgender judgement and that they would change how they speak to be more feminine around cis listeners – evidencing mental salience of the cisgender listening subject – only two out of the four speakers implement this change in practice with /s/. Future research considering multiple linguistic features together could be especially illuminating here; it is possible that the Alpaca and Squirrel do also make changes towards more hegemonically feminine indexicalities with the cisgender listener, but that these changes are along features other than /s/, such as overall pitch (f0), intonation, voice quality, or discourse markers. I return to the issue of investigating only one linguistic variable below.

In the model I initially constructed, speakers who expressed greater degrees of struggling with voice dysphoria, greater desire have a voice which 'passes' as the voice of a cisgender woman, or a greater sense of danger and pressure to conform around cisgender listeners would be more likely to shift to a fronter /s/ pronunciation. On the other hand, speakers who expressed stronger attitudes of resistance against cisnormativity, differentiated themselves from hegemonic femininity, or were concerned that their voice might make them sound like a gay man would be more likely not to shift to a fronter /s/ pronunciation. In interpreting the explanatory failure of this model on my data, it is first important to note that its formulation tacitly constructs a mutual exclusivity of the former and latter concerns, which is not upheld by my data. It seems that for Mantis and Jellyfish, the

former concerns (which I thought would predict a shift to fronter /s/) are caused by the latter concerns (which I thought would predict no shift) – it was discomfort with hegemonic feminine norms and concerns with gay-male misrecognition which caused these speakers to feel pressure to conform around cisgender listeners. In the discussion below, I return to the literature to reconsider the predictive factors I initially proposed in light of the patterns which actually emerged, specifically focusing on interpreting my results through the lens of indexicality.

The issue of misrecognition is central here. In Mantis' discussion of how her voice is percieved, sounding like a straight man is not considered a possibility; if she is perceived correctly she will sound like a queer woman, and if she is perceived incorrectly she will sound like a gay man. Within the model of /s/ indexicality brought forth by Calder (2019a) (recall figure 1.1), other semiotic information is used to determine through which indexical order /s/ pronunciations will be interpreted. From this perspective, a shift in /s/ pronunciations could be considered useless for Mantis, in that whether she will be heard as a gay man or a cis woman cannot be determined by her /s/ pronunciation. Yet she shifts, contradicting the model. The extent of her shift provides help here; she shifts to /s/ pronunciations which fall in closer line with those expected of cis women than those reported for cis gay men. Within the literature reviewed, Zimman (2013) gives the highest mean CoG for cisgender gay men (not in drag) in the literature reviewed at 6550Hz, whereas Mantis' /s/ pronunciations with the Cis CP had a mean CoG of 7964Hz, well within the hegemonically expected range of cis women. In Calder (2019a), SoMa queens front /s/ further than Castro queens in order to forge a higher indexical order and index a radical queer femininity beyond mainstream cisgender gay male femininity. Here, I argue that Mantis fronts /s/ further than cis gay men do, but that no higher order of indexicality can be forged, as she must make do with the two hegemonic indexical orders already existing in the assumed indexical inventory of her listener; instead, she uses this much more fronted hegemonically-feminine /s/ access to the lower level of indexicality and encourages her /s/ pronunciations to be interpreted as those of a woman on the nth level, and not those of a gay man on the $n + 1st$.

Although this extension of the model works for Mantis, the situation is different for Jellyfish. If it is indeed the case, as Jellyfish mentions, that "when you're a lesbian sounding like a gay guy and like having a faggcent will get you clocked way more than sounding like boy," why does she choose to shift to a fronter /s/ rather than not shifting /s/ and 'sounding like a boy'? Although it is not easy to answer this question with certainty from the data available and without context of other indexical features, I see three possible explanations. First, it is possible that Jellyfish is simply mirroring the /s/ pronunciations of her CPs, and that questions of perception of gender and sexuality are less relevant to this case. A point in favor of this explanation is the priority Jellyfish lends to her lesbian identity; because the Cis CP was also a lesbian, it is possible that the function of linguistic mirroring in this case could have been a move to create greater similarity between herself and her CP, a move of 'adequation' from the perspective of the 'tactics of intersubjectivity' theorized by Irvine and Gal (2000). The second possible explanation

has to do with the complexity of Jellyfish's relationship with expressing boyness. Although she indicates that it is at least hypothetically safer for her to sound like a boy than a gay man around cis women, she also clarifies that "I'm like less concerned about like trying to like overly feminize my voice when I'm around other trans people", and that "I love talking like a boy when I'm interacting with other boys". From this perspective, it is possible that she felt less comfortable to sound like a boy with the Cis CP, and thus "with like, it's mostly people I don't know, I'll like try harder to um have like a more conventionally feminine voice." A point of dissatisfaction with this second explanation is that it breaks down when considered for the other speakers; as mentioned above, all speakers indicated similar sentiments that they would shift towards hegemonically feminine speech with cisgender listeners, but Alpaca and Squirrel do not in fact shift /s/ in this way. The third, and perhaps more promisingly generalizable, explanation for Jellyfish's shift has to do with agency and the notion of a 'natural' voice. In her discussion of her concerns with misrecognition, she underlines that although she is proud of "being able to more or less manipulate [her voice] how [she] want[s]", she nonetheless is sometimes bothered that she might

> "default back to talking like a gay guy around cis women ... It's kind of rolling the dice what voice comes out on a certain day, and sometimes the faggcent comes back"

Her notion that a voice more prone to misrecognition ('the faggcent') is a 'default' for her around cis women and her metaphor of styleshifting to rolling dice make it clear that she does not feel fully in control of her voice. Here, it is possible to understand her shift to a fronter /s/ pronunciation with the Cis CP as an example of this sort of 'defaulting', rather than an agentive choice aligned with her intentions and goals. It is further interesting that in her discussion of how she talks around other trans people, comfort in her voice is also described in terms of defaulting:

> "at a certain point, I like stop performing and sort of default back to like my neutral state which is still just like talking like a boy, which is fine, like I don't really feel much pressure to change that."

Although Jellyfish does acknowledge that "it's all performance", in line with feminist understandings of gender performativity, it is nonetheless notable that Jellyfish perhaps considers hegemonic femininity somewhat of a difficult or unnatural performance for her, and that it does not feel like her neutral state. Here, it is clear that styleshifting is not effortless, always available, and without cost; instead, it comes to be limited.

From this, I am motivated to take up the question of the interplay between agency and limits on agency as central to the interpretation of my data. In the introduction to this thesis, I mentioned that agentic models of linguistic variation, which see speakers' linguistic practices as strategically and intentionally deployed to achieve social goals, was limited by the restrictions imposed by the importance of the listener's role in interpretation. From my results, it seems additionally im-

portant to consider the degree to which variants are available and natural to speakers; even if a speaker's vocal tract holds the potential to produce a wide degree of linguistic variants, it is nonetheless possible that a speaker may only be familiar with using only a smaller range of variants, or experience some variants as more effortful and less natural due to their varying familiarity. An analogy can be made to phonology in language learning; it is an uncontroversial fact of language that all humans with typical vocal tracts are capable of all phonetic and phonemic distinctions in all human languages, but it is also uncontroversial that learning the phonology of a new language is challenging and time-consuming, and that effort is required to sustain accurate pronunciation.

I argue that this framework of limited agency helps explain Alpaca and Squirrel's lack of normative shifting as well. For Squirrel, the reasoning is fairly simple; as a transfem early in the process of transition, Squirrel considers a feminized voice as only accessible in the future. The inaccessibility of her desired voice – her voice not being "they way [she] would want it to be if [she] could like, pick it" – is complete enough that she does not use her voice at all on the internet, the sphere in which she wishes to be perceived as feminine. Being not fully out in in-person interactions means that she is for the time being continuing to assume she might be perceived as masculine, and does not yet feel able to combat these perceptions through the voice.

For Alpaca, I believe that this framework can help explain her lack of a frontwards shift, although the backwards directionality of her shift remains difficult to explain. Alpaca mentions learning "the exercise to raise your larynx", and has the sense that she has implemented this into her everyday speech over the course of her transition, noting that she "learned how to do it and then [she] found [her]self sort of subconsciously doing it" over time. The status of agency in this example is instructive to unpack; despite the ability to raise the larynx always existing in her vocal tract, learning the exercise she mentions provides a greater degree of access to this variant. That she found herself "subconsciously doing it" indicates another complex relationship with agency, whereby changes could not be forced but become more natural and less effortful over time. With respect to /s/, it is possible that this resource for gendered indexicality is not easily accessible for Alpaca, and perhaps feels less needed in the context of her perceived success in feminizing her voice over the years of her transition. Because shifting /s/ fronter would require accessibility and effort, this change is not made. That this cannot account for why Alpaca shifts to backer /s/ pronunciations is not ideal, but perhaps not a problem for the framework; there do not seem to be any factors found in Alpaca's identity or interview which could be used to explain the directionality of her shift, so it is likely that more data would be needed to resolve this issue.

It is also important to clarify that this framework of limitations on agency certainly does not imply that all of my speakers' behavior can only be explained by limitations rather than agentivity. As discussed above, I argue that Mantis represents an example of agency as a better explanation than limitation, in that her shift is the most drastic, and it simultaneously achieves both of her goals; to protect herself from dysphoria and negative judgment by her listener by taking up fronted

/s/ as a hegemonically feminine index, and to protect herself from gay-male misrecognition by using fronter /s/ pronunciations than would be expected from gay men. My first possible explanation of Jellyfish's behavior may also suggest agency as a primary explanatory frame. These sorts of goal-satisfying use of linguistic resources is unproblematically read as agentive, even though it is defensive in Mantis' case. My argument here is for the both-and of agentivity and limitation, seeing that my data is best explained when both considerations are given weight and relevance. After all, this model more suitably describes the social world my participants live within; they are limited by their societies and their abilites, but nonetheless can agentically and strategically make choices when those choices are accessible and intentionally available.

Conclusion

In conducting an initial investigation transfeminine speakers' negotiation of cisheteronormativity through the use of the /s/ sound, this thesis aims to shed light on the uneasy interplay of the limitations that power structures impose and the agency individuals have to resist those norms. Research on trans speech has primarily placed its focus on one or the other side of this contrast. When speakers are investigated in only one context, whether it be a safe queer space, or a possibly-unsafe cisgender space, the contrasting influences of safety and conformity on behavior cannot be seen clearly. As a particularly complex site of gendered speech performativity, fronting /s/ is a difficult choice for transfems; on the one hand, survival within the cisgender world is often difficult and requires conformity, but on the other, conformance requires effort, is not always readily available even if physically possible, and should be resisted. Furthermore, even if a speaker does set out to conform to the hegemonic expectations of feminine speech by fronting /s/, it is not guaranteed that this will gain them the acceptance they seek; the risk remains that a cisgender listener may (intentionally or otherwise) misrecognize this index to signal a gay-masculine identity. In this way, they accept queerness but reject transness, protecting themselves in the comfortable misperception of transfeminine people as cisgender gay men.

The difficulty of sociolinguistic decisions is underlined by the unexpected correspondences between my speakers' /s/ shifting behavior and their expressed attitudes. The particularly vexing conflict between the indexical orders of /s/ and the threat of misrecognition this produces for transfeminine speakers attests to the analytic value of misrecognition as theorized by Barrett and Hall 2024 as a framework to understand the interaction of power structures and everyday behavior. Traditional sociolinguistic frameworks would predict that speakers who wished to differentiate themselves from hegemonic norms and present their desired identity would simply do so, where my data shows that this ability is sometimes stripped from speakers, who may have to prioritize their own feelings of safety and comfort over their identitymaking goals, or who may lack knowledge or practice with the indexical materials they would need to do so.

In proposing a nuanced approach to sociolinguistic decisions, this thesis does not intend to argue that marginalized speakers are stripped of agency and have no choice about how they present their voice, simply following norms to the extent they are able. This approach would fly in the face of the decades of sociolinguistic research which have worked to uncover the agentic freedom that linguistic vari-

ation offers in allowing speakers to present themselves and their voices in line with their social goals. Instead, I argue that agentic goals and limiting restrictions are always simultaneously present, and speakers, especially marginalized speakers such as transfeminine people, make difficult decisions in consideration of all of the factors. Importantly, speakers can tip the scales in the favor of agentivity if they are empowered with semiotic self-awareness; as my participant Jellyfish says,

> "People take this shit way too seriously ... Fundamentally, queerness should change how you move through the world. I think that's a good thing, I think like having control of how you're perceived and how you interact with the world and thinking about that more is something that everyone should do, and my transness and my lesbianism are like vectors to which I do that."

5.1 Limitations and Directions for Future Research

A primary limitation of this study is its lack of a raciolinguistically informed approach. As a white researcher working with mostly white participants and drawing on a mostly white-centric research literature, I do not aim for this study to perpetuate the whiteness of trans studies, queer studies, and sociolinguistics. Research on BIPoC transfems' /s/ pronunciations on their own terms and in their own spaces would be highly illuminating. J. Calder and King (2022) find that a gender split in /s/ pronunciation is not present for Black speakers in Bakersfield, California. If a study like this one were conducted with Black transfeminine speakers, it is possible that race of the perceiver may emerge as equally relevant as cisgender status, and that the influence of the white cisgender listening subject would exert multiplicative oppressive effects, an example of sociolinguistic intersectionality (see miles-hercules 2022). Due to the dearth of research on Asian-Americans in the sociolinguistic literature, and the absence of research on Asian-American /s/ pronunciation, It was regrettably difficult for me to determine the extent to which the whiteness of the hegemonic gendered expectations of /s/ frontness impacted the Asian-American speakers in my study, Mantis and the Trans CP. I encourage further research outside white speaker and perceiver populations, and hope to be able to work with queer and trans BIPoC sociolinguists and speakers to focus closer on race in the future.

Other limitations of this study pertain to scale and scope. As mentioned in my discussion, this study could greatly benefit from more angles of analysis in several ways. To name two, expanding to include multiple linguistic variables alongside /s/ would provide much-needed context and insights, and zooming in to the discourse level to inspect how /s/ was shifted within each individual conversation based on topic, stance, and mood would allow a more nuanced understanding of the moment-to-moment meaning-in-context of /s/ as it develops and shifts in conversation with the aggregated understanding I present. Further research would also benefit significantly from a larger scale, at which a greater diversity of participants and conversation partners could reveal broader patterns and reduce the

possible confounds introduced by any idiosyncrasies of the individual personalities of the two conversation partners I have. This would also allow for a quantitative analysis of predictive factors of /s/ shifting, operationalizing with more precision the attitudes, concerns, and identities I consider qualitatively here.

Appendix A

Survey

Appendix B

Interview Questions

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